Life Cycle Assessment

Environmental assessment of Graphite Electrode



Graphite India Limited

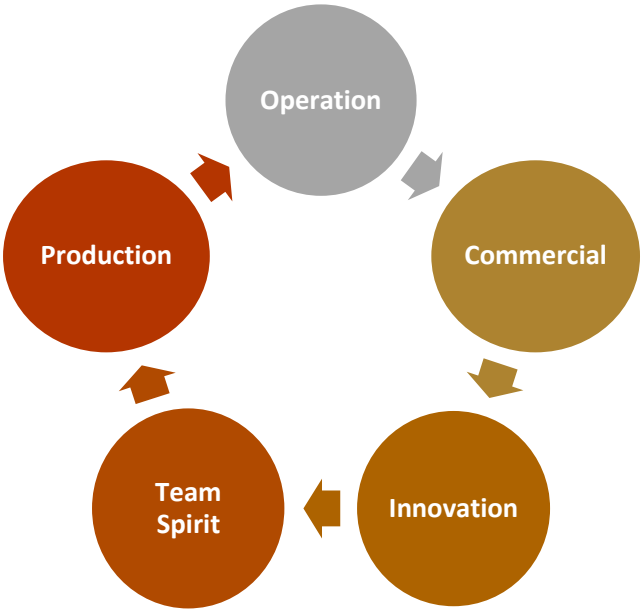
April 2024 | LCA

Contents

Description	Page#
About Graphite India Limited	3
Executive Summary	4
About Life Cycle Assessment	5
Goal and Scope Definition	6-8
Software and Data Sources	9
Life Cycle Inventory Data	10-12
Inventory Analysis	13-16
Impact Assessment Method	17
Impact Categories	18-19
Life Cycle Interpretation	20
Overview of Environmental Impacts	21-33
LCA Conclusions	34
Appendices	36-41

About Graphite India Limited

Graphite India Limited (GIL) is the pioneer in India for manufacture of Graphite Electrodes as well as Carbon and Graphite Speciality products. GIL's manufacturing facilities are spread across several plants in India and it has also got a 100% owned subsidiary at Nuremberg, Germany, by name Graphite COVA GmbH. Being a responsible corporate citizen, GIL is on a business growth trajectory following a journey of sustainability. With over 50 years of experience, GIL's excellence focuses on five areas



Graphite India Ltd (GIL) is the pioneer in Carbon and Graphite products in India. It came into existence way back in 1960s. Starting with Graphite Electrodes, the core product of the company, it has entered into the coveted field of Speciality application of carbon and graphite. By further leveraging the core competence and economies of scale GIL intends to advance the production capacity cost efficiently. With efficient global partners and proximity of the manufacturing facilities to inland customers and seaports, GIL is able to ensure timely deliveries to its customers. Over the years GIL has emerged to become an established and reputed brand in the Global Market. The corporate philosophy of the company lays emphasis on consistent quality of products and services across all its divisions which have culminated to an enviable track record in the industry arena. Having started in 1967 in collaboration with erstwhile Great Lakes Carbon Corporation (GLCC) of USA, GIL has been continually improving its product quality and services thereby scaling newer heights of excellence and customer recognition. This journey has been fueled by GIL's reliance on cutting-edge technology, a natural penchant for innovation and creativity, eco-friendly approach in production process, consistency of product quality and services as well as productivity and cost optimization.

25+ **Awards Recognitions**

50+ **Years of Excellence**

Executive Summary

This report is a convergence of different operational aspects of Graphite Electrodes to meet the sustainability agenda and product standards. A life cycle assessment (LCA) study was performed for the main product, graphite electrode, to assess the environmental performance and product footprint. The study covers the Cradle to Gate approach by ensembled different life cycle stages i.e. raw material extraction, upstream transportation of raw material and fuel, manufacturing and finishing of the product till manufacturing plants' out-gate.

Graphite Electrodes are manufactured in the Durgapur and Satpur factories. These two were chosen as the study area for life cycle inventory analysis. Graphite electrode is an essential requirement for electric arc furnaces (EAF) primarily into steel production from scrap to reduce the CO_2-emissions. With the growing consciousness of the world towards a low carbon future, a paradigm shift is of utmost importance considering sustainable resource and energy consumption.

The study considers the raw material extraction stage, upstream transportation of raw materials and fuels, and the manufacturing which includes consideration of energy inflow and outflow through emissions at different life cycle stages followed by finishing. The usage stage and end of life treatment of the product considered is out of the scope of this study.

The CML method has been chosen as the main environmental impact assessment method for this study, and the selected indicators for this study are based on EN 15804 (EN15804, 2012+A1_2013) and EN 15978 (EN15978, 2011), which are: Abiotic Depletion, Acidification Potential, Eutrophication Potential, Freshwater Aquatic Ecotoxicity Potential, Global Warming Potential, Global Warming Potential excluding biogenic carbon, Photochemical Ozone Creation Potential, Ozone Layer Depletion Potential and Primary energy demand from renewable and non-renewable resources.

The result indicates a net 118.78% of resource saving potential, which is the positive attributional impact element in regard to the sustainable consumption of waste product. However, the carbon footprint for total raw material extraction life cycle stage is found to be 0.828 kg CO_2 equivalent for 1 kg graphite electrode production. The usage of CPC, which is a petroleum byproduct tends to net savings in ADP for this stage which envisage sustainable consumption. Besides, an understanding of the significant share of impact indicators, such as GWP, POFP, ODP, HTP, MAETP, FAETP, etc., is provided by grid electricity consumption. This provides a framework for methods aimed at minimizing the use of non-renewable energy sources.

During this life cycle evaluation, the sustainable use of Coal Bed Methane and Natural Gas as a fuel prevented the depletion of natural resources like methane as well as a massive quantity of emissions that have the potential to have an even greater negative impact on global warming than CO2. In terms of GWP potential, over 68% of GHG emissions have been reduced, and in terms of photochemical oxidant generation, 45.72%. These results provide insight into the sustainability of the product and align with the global climate risk mitigation plan aimed at achieving net zero targets.

About Life Cycle Assessment (LCA)

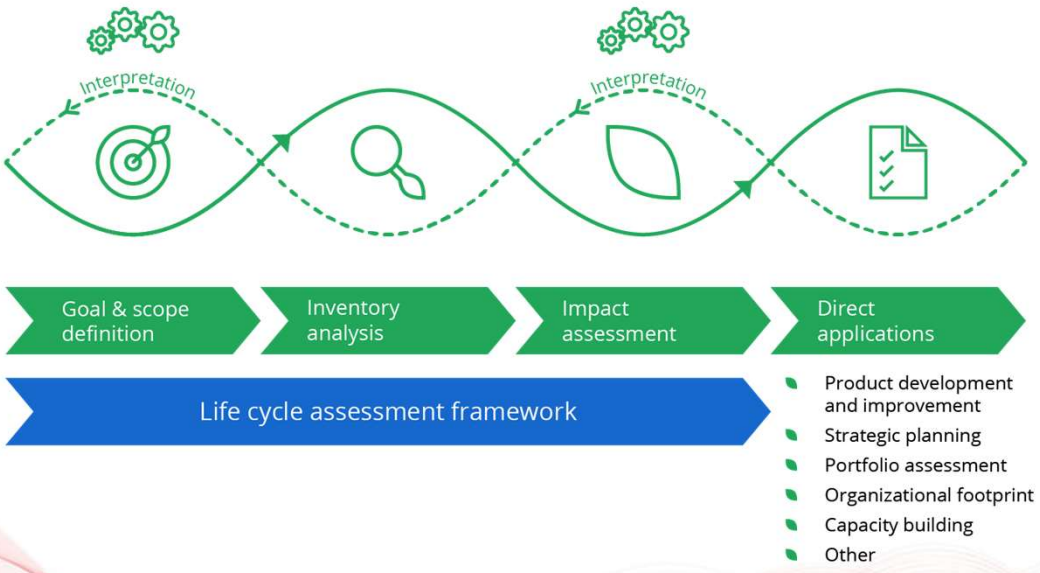
An essential technique for assessing the environmental impacts of a process, product, or service throughout the course of its full life cycle is life cycle assessment, often known as life cycle analysis. This covers the broader steps of raw material extraction, manufacture, usage, and post-use considering the system approach. These life cycle stages can all have different effects on the environment. Businesses can create more sustainable approaches to product development and strategic planning by adopting Life Cycle Assessment (LCA), which gives them priceless insights into the environmental effects of their goods and services.



The use of LCA has several advantages. Starting with product development, environmental communication, strategic planning, and evidence-based policymaking can all benefit from the insights gathered from an LCA. Product designers, for instance, can investigate how their decisions impact the items' sustainability. Decisions by policymakers can be made by weighing all significant environmental effects. Managers of sustainability can evaluate the portfolio and determine what is required to meet carbon footprint targets. Marketing teams have access to factual information for communications on sustainability. Which suppliers have the most environmentally friendly goods and practices can be found out by the purchasing department.

LCA is a standardized methodology, which makes it reliable and transparent. The International Organization for Standardization (ISO) provides standards for LCA in ISO 14040 and 14044. These standards describe the four main phases of an LCA:

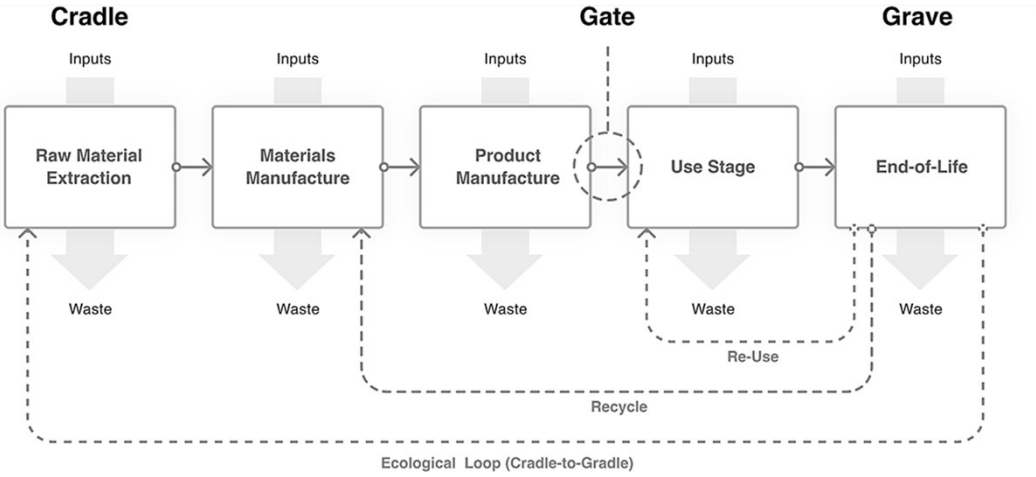
1. Goal and scope definition
2. Inventory analysis
3. Impact assessment
4. Interpretation



Goal & Scope Definition

The goal of the study is to carry out a life cycle environmental evaluation of graphite electrode production. This report acts as an initial assessment, to provide a first indication for the product sustainability standard, based on generic data. Further optimization, using more specific LCI data and strategic enforcement in terms of energy usage and raw material acquisitions, the LCA must be carried out. This is particularly important when taking into account the product standard considering its environmental performance for better decision making to achieve the net zero goals.

This study aims to identify a first indication of the environmental performance of graphite electrode productions from the cradle to gate life cycle perspective. Thus, the results represent intermediate results in the planning process and therefore no comparative assertion.



System Boundary

This section outlines the life cycle stages of the studied systems, which constitute the system boundaries. The system boundaries shall be consistent with the goal and scope of the study and determine the processes and flows included in the LCA of Graphite Electrode. It defines the scope of the system and is subsequently examined. The system boundaries also define what is not part of the study because it is outside the system.

Establishing system boundaries define the stages, processes and flows that are taken into account in the LCA. These should comprise (a) Actions necessary to accomplish the goals of the current study and, as a result, to perform the functions under consideration and (b) Processes and flows that significantly increase the possible effects on the environment and the economy.

The system boundaries of the process flow covers the value chain, including raw material extraction, raw material transportation, fuel and electricity consumption, processing and finishing so-called from “cradle-to-gate”. This can be represented as a partial product supply chain, from the raw materials (cradle) up to the manufacturer’s “gate”. The usage phase of the graphite electrodes and further waste treatment stage are outside the system boundaries. A graphical scheme of the system boundaries is shown below.

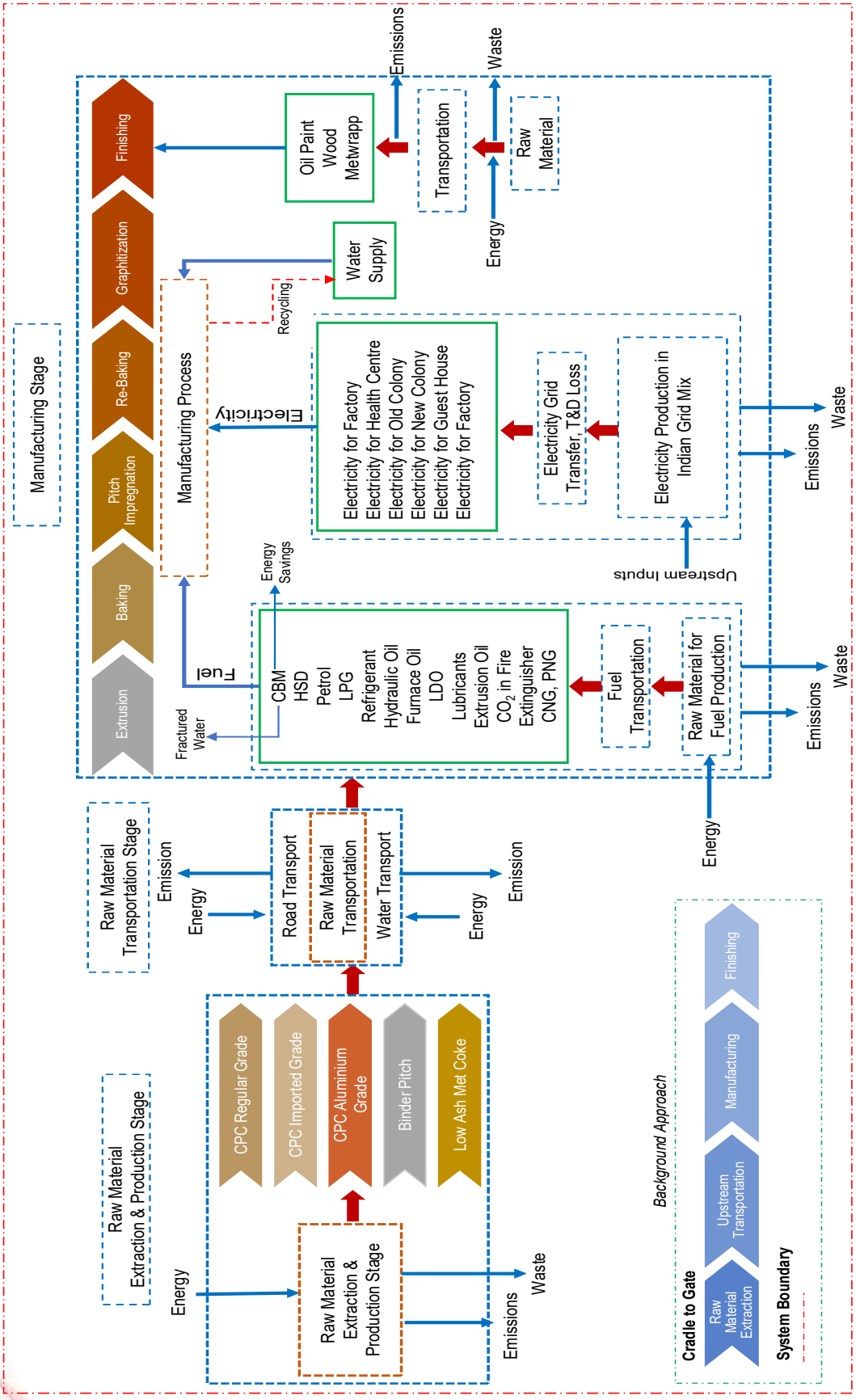


Figure: Schematic Diagram of "Cradle to Gate" System Boundary of Graphite Electrode

Assumptions & Limitations

Consideration of system boundaries requires certain assumptions and limitations. In order to assess the different product stage environmental performances, definite boundaries need to be selected for an in depth assessment. As graphite electrode is the primary product of GIL initiating from the raw material exploration to its production, the study contemplates the "Cradle to Gate" approach and ostracize the usage phase in EAF followed by the end of life treatment. This study focuses on the production scenario of Durgapur, West Bengal GE division and Satpur, Nasik GE division as closely as possible. The following assumptions have been considered:

For the raw material stage, the production of raw materials, including its upstream transportation, energy usage, emissions, and waste generation, has been contemplated. Calcined Pet Coke (CPC) is a by-product of petroleum refinery operations, whereas Low Ash Met Coke (LAMC) is considered to be coking coal. The raw materials contributing 90-95% of the physical allocation for the manufacturing of graphite electrode were taken into consideration. These are Calcined Pet Coke (CPC) Regular Grade, Calcined Pet Coke (CPC) Imported Grade, Calcined Pet Coke (CPC) Aluminium Grade, Pitch Binder (Tar), and Low Ash Metallurgical Coke (LAMC).

For the raw material and fuel transportation stage, only land and water transport mediums have been taken into account for the life cycle model.

The third stage, the manufacturing of graphite electrodes has certain unit operations, which are Extrusion, Baking, Pitch Impregnation, Re-baking, Graphitization and Finishing. For these functional units, energy and water inputs have been considered. Besides, the life cycle impacts of key attributes like energy and water have been included within the boundary. Besides, energy like electricity and fuel utilised by the whole GIL, Durgapur and Satpur plant activity for product development has been included within the system boundary. The fuels under consideration are Coal Bed Methane (CBM), CNG, PNG, HSD, Petrol, LPG, Refrigerant, Hydraulic Oil, Furnace Oil, LDO, Lubricants, Extrusion Oil, CO_2 in Fire Extinguisher. CBM, which is basically methane gas found in the coal seams, is extracted and stored for energy generation. Water requirements for CBM fracturing as well as fugitive emissions during CBM production have been contemplated within the system boundary. The materials used for the packaging and finishing are Metwrapp, Oil paint and Wood which are accounted for within the boundary considered.

Graphite electrodes are used in secondary steel production in Electric Arc Furnace (EAF) in steel plants using scrap steel. But for analysing the environmental performance of the product, the downstream distribution, usage, and waste treatment after end of life phase have not been taken into consideration within the system boundary.

Software & Data Source

For assessing the environmental performance of graphite electrode, life cycle impact assessment (LCIA) tool have been used. The tool is modelled in commercial software OpenLCA, using the appropriate life cycle inventory (LCI) data for energy, materials and transport activities. A critical review of the system boundary, model and methodology has been carried out to ensure the quality of the LCA tool, in line with ISO 14040: 2006 and ISO 14044: 2006 standards (14040, 2006) (14044, 2006). OpenLCA version 2.0 has been used together with ecoinvent database version 3.9.1. Regional data for the key materials has been used. Considering the geographical location of this case study, Electricity data is used for the electricity grid from West Bengal, India. European data is taken as the next best available data and where this is not possible, global data has been used. This is therefore a useful study during the design phase, showing the relative order of magnitude for each life cycle stage. The data is selected based on the technology representativeness and the data availability from the GIL, Durgapur and Satpur plant and detailed information on the source and representation of the data is provided in following Tables.

Life Cycle Inventory Data

Raw Material

Raw Materials	Consumption per kg GE	Unit
CPC, Regular Grade	0.290	kg/kg GE
CPC, Imported Grade	0.359	kg/kg GE
CPC, Al grade	0.078	kg/kg GE
Binder Pitch	0.246	kg/kg GE
Pitch_PI	0.228	kg/kg GE
LAMC	0.290	kg/kg GE

Raw Material Transport

Raw Material Transport	Mass*Distance	Unit
Transport of CPC Regular Grade, Road	0.463	t.km/kg
Transport of CPC Imported Grade, Road	0.066	t.km/kg
Transport of CPC Al Grade, Road	0.036	t.km/kg
Transport of LAMC Met coke, Road	0.209	t.km/kg
Transport of Binder Pitch, Road	0.110	t.km/kg
Transport of CPC Imported Grade, Sea Freight	2.549	ship t.miles/kg

Fuel Transport

Fuel Transport	Mass*Distance	Unit
Transport of Low Sulfur Heavy Stock, Road	0.00089	t.km/kg
Transport of LDO, Road	0.00059	t.km/kg
Transport of Extrusion Oil, Road	0.00084	t.km/kg
Transport of Diesel, Road	0.00031	t.km/kg
Transport of LPG, Road	2.95107E-08	t.km/kg

Electricity Consumption

Electricity	Electricity Consumed	Unit
Electricity Factory	3443.544	kWh/MT
Electricity from grid for Health Centre usage	0.034	kWh/MT
Electricity from grid for Old colony	1.042	kWh/MT
Electricity from grid for New colony	0.488	kWh/MT
Electricity from grid for guest house	0.134	kWh/MT
Total Electricity	3.445	kWh/kg

Life Cycle Inventory Data

Fuel Consumption (Miscellaneous)

Description	Consumption	Unit
Petrol in own vehicles	6.08793E-05	kg/kg GE
LPG in site	0.000005	kg/kg GE
LPG in Office (canteen, pantry, temple,lab)	0.0000055	kg/kg GE
LPG at Guest House	0.0000030	kg/kg GE
LDO (Light Diesel Oil)	0.0000000	kg/kg GE
Furnace Oil	0.0000000	kg/kg GE
CO2 in process during Project	0.0000000	kg/kg GE
Lubricants	0.0006666	kg/kg GE
Extrusion oil (lub oil)	0.0022	kg/kg GE
Wood and paper in ACHESON and LWG	0.0000	m3/kg GE
CO2 fire extingushers	0.0000027	kg/kg GE
Refrigerant (type R-22)	0.00000026	kg/kg GE
Refrigerant (type R-410)	0.00000015	kg/kg GE
Refrigerant (type R-407)	0.00000034	kg/kg GE
Refrigerant (type R-32)	0.00000041	kg/kg GE
Refrigerant (R-22) GH	0.00000005	kg/kg GE
Low Sulphur Heavy Stock (LSHS)	0.0024	kg/kg GE
Hydraulic Oil	0.0005	kg/kg GE
Low Sulphur Heavy Stock (LSHS)	0.003286	kg/kg GE
LDO/ Indus.Diesel Oil	0.002190	kg/kg GE
Extrusion Oil	0.002979	kg/kg GE
Furnace Oil	0.000000	kg/kg GE
LPG	0.000005	kg/kg GE

Description	Consumption	Unit
LAMC carbon and VMC burning	0.0932	kg/kg GE
AL grade CPC carbon and VMC burning	0.0643	kg/kg GE
Breeze carbon burning and VMC burning	0.0086	kg/kg GE
Saw dust burning	0.0014	kg/kg GE
Binder Pitch volatile loss	0.0535	kg/kg GE
Impregnation pitch volatile loss	0.0528	kg/kg GE
R-22	0.000000	kg/kg GE
R-32	0.000001	kg/kg GE
R-407	0.000002	kg/kg GE
R-134	0.000000	kg/kg GE
LAMC Carbon and VMC Burning	0.124355	kg/kg GE
Binder Pitch volatile loss	0.078010	kg/kg GE
Impregnation Pitch Volatile Loss	0.042691	kg/kg GE
BMC (Cutting Gas)	0.000084	kg/kg GE
CNG	0.010462	kg/kg GE

Life Cycle Inventory Data

Fuel Consumption (Diesel)

Fuel Description	Consumption	Unit
Diesel in own/controlled vehicles	0.000472976	kg/kg GE
Diesel in own/controlled vechiles in project work	2.63324E-05	kg/kg GE
Diesel in own/controlled vehicles in ICD ware house	2.86385E-05	kg/kg GE
Diesel in DG Set	0.000440587	kg/kg GE
Diesel in Machinery	0.001858112	kg/kg GE
Diesel	0.002296677	kg/kg GE

Fuel Consumption (Coal Bed Methane, CBM)

Fuel Description (CBM)	Quantity	Unit
CBM Combustion	0.438	kg CO2/kg GE
CBM Water for Fracturing	0.290	kg CBM water/kg GE
Methane saving in terms of CBM utilisation	0.1590	kg CBM/kg GE
Methane, Fugitive Emission	0.0250	kg/kg GE
Carbon Dioxide, Fugitive Emission	0.1170	kg/kg GE

Water Consumption

Water Consumption	Water Consumed	Unit
Water	1.071	l/kg GE

Packaging & Finishing Goods

Packaging & Finishing Goods	Consumption	Unit
Oil Paint	0.00002	kg/kg GE
Wood consumption	0.0000071	m3/kg GE
Met wrap	0.008	kg/kg GE

Output Flows

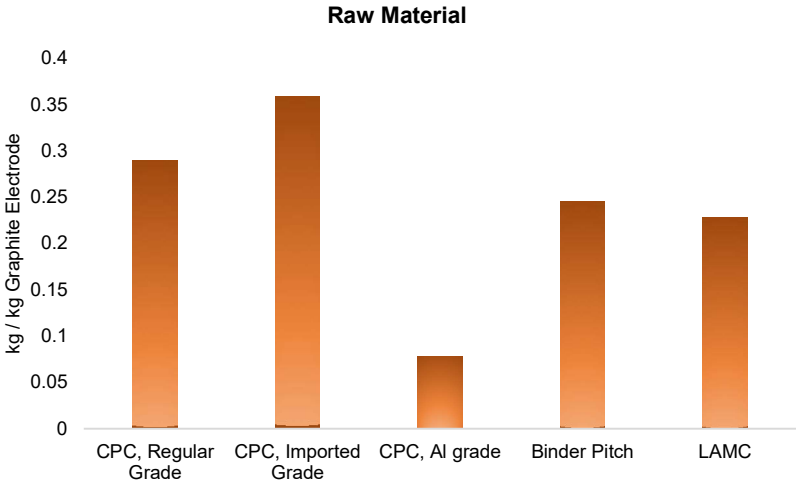
Output Flows	Amount	Unit	Description
Graphite Electrode	1	kg	Output Product
Carbon dioxide, fossil	0.117	kg	Fugitive emission during CBM extraction
Methane, fossil	-0.159	kg	Methane entrapped as CBM
Methane, fossil	0.025	kg	Fugitive emission during CBM extraction
Carbon dioxide, fossil	0.438	kg	CBM Combustion

Inventory Analysis

The Life Cycle Inventory (LCI) of graphite electrode production is a set of multiple stages involved from raw material extraction, upstream transportation, manufacturing of electrode using resources, energy and water as required. For the ease of analysis, the whole inventory has been distributed into seven categories. These are raw materials, raw material transportation, electricity, fuel, water, upstream fuel transportation and packaging & finishing. The inventory was prepared for 1kg of graphite electrode production from GE division in Durgapur and Satpur. Categorical inventories are described as follows:

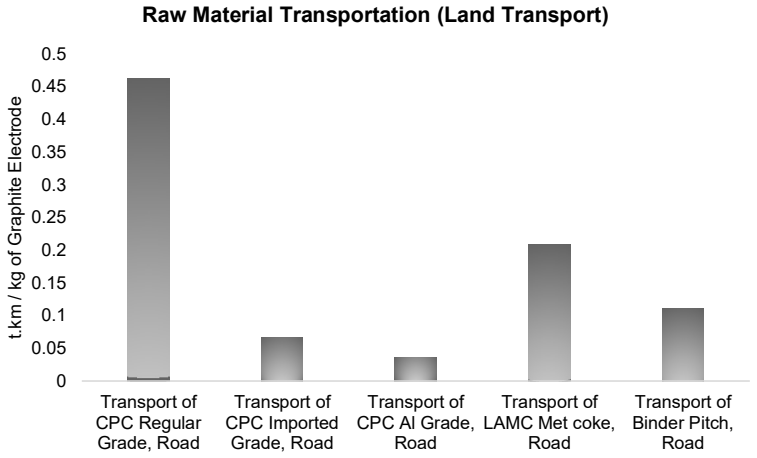
Raw Material

The primary raw materials for graphite electrode production are calcined petroleum coke which is petroleum refinery waste product to utilize. For the life cycle inventory, primary raw materials were taken into account based on their percentage allocation for FY 2023-24 from Durgapur and Satpur division. 90-95% of the raw materials are carbon based product among which regular grade calcined pet coke (CPC) is 24%, imported grade calcined pet coke is 30%, low ash metallurgical coke (LAMC) is 19% and binder & impregnation pitch is around 20% respectively. For LCI dataset, the share of raw materials is presented for 1 kg of graphite electrode production in the following figure.



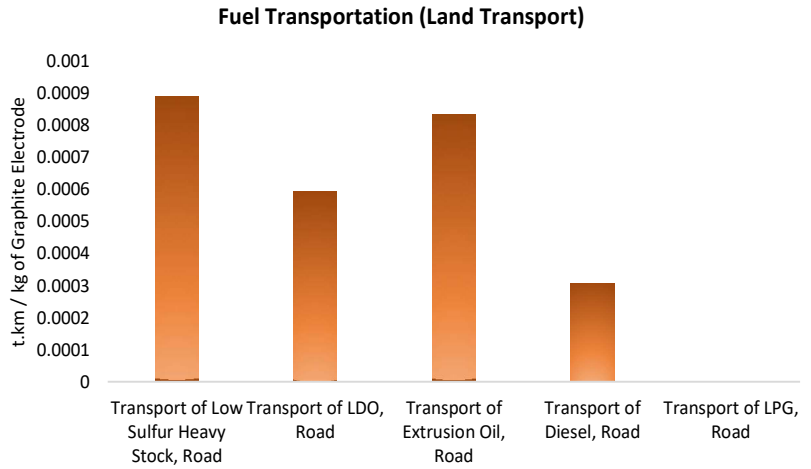
Raw Material Transportation

In LCI preparation, all the raw materials i.e. CPC regular grade, CPC Al grade, Binder Pitch, and LAMC are transported through land transport except CPC imported grade, which relies on water transport too. 52.3% share of land transport for CPC regular grade being highest, 23.62% share of land transport through road is for LAMC . Whereas, 4.1% is for road transport of CPC, Al grade, 7.5% for CPC imported grade and 12.45% for binder pitch transportation. However, 2.548 ship t.miles of water transport has been accounted for CPC imported grade for production of 1 kg of graphite electrode. For LCI dataset, the share of raw material transportation is presented for 1 kg of graphite electrode production in the following figure.



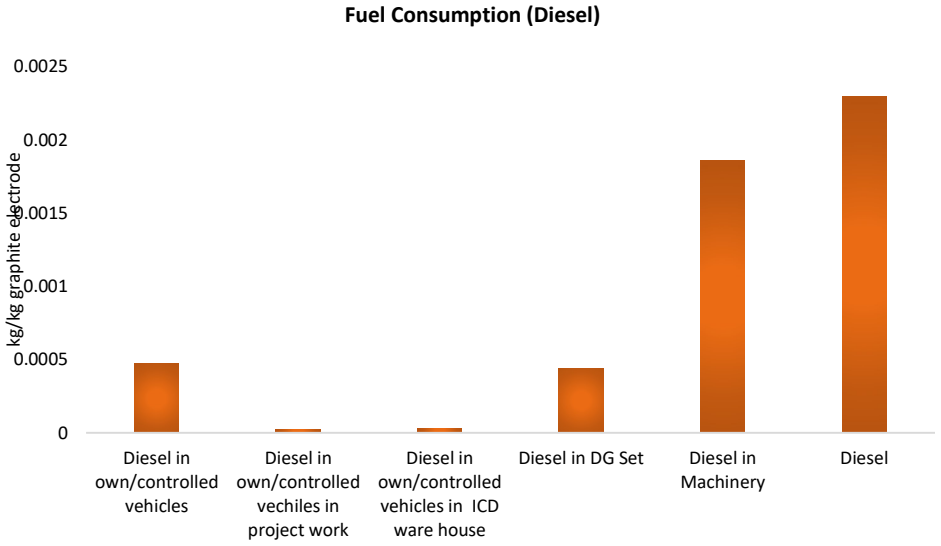
Upstream Transportation of Fuel

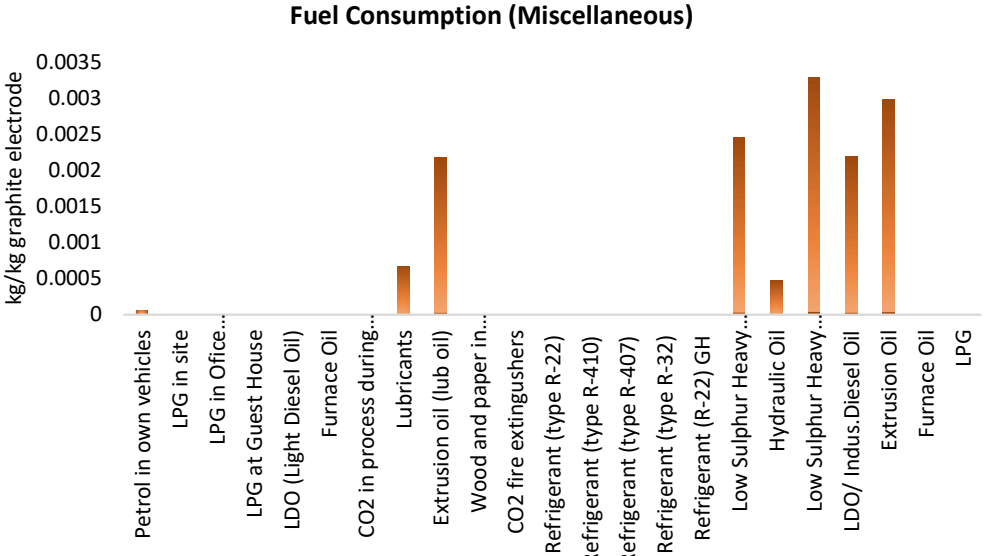
For upstream fuel transportation data inventory, low sulfur heavy stock (LSHS), light diesel oil (LDO), extrusion oil, diesel and LPG have been considered through road transport. The unit data has been depicted in the following figure. 99% share of upstream fuel transportation is based on LSHS, LDO and Extrusion oil transport.

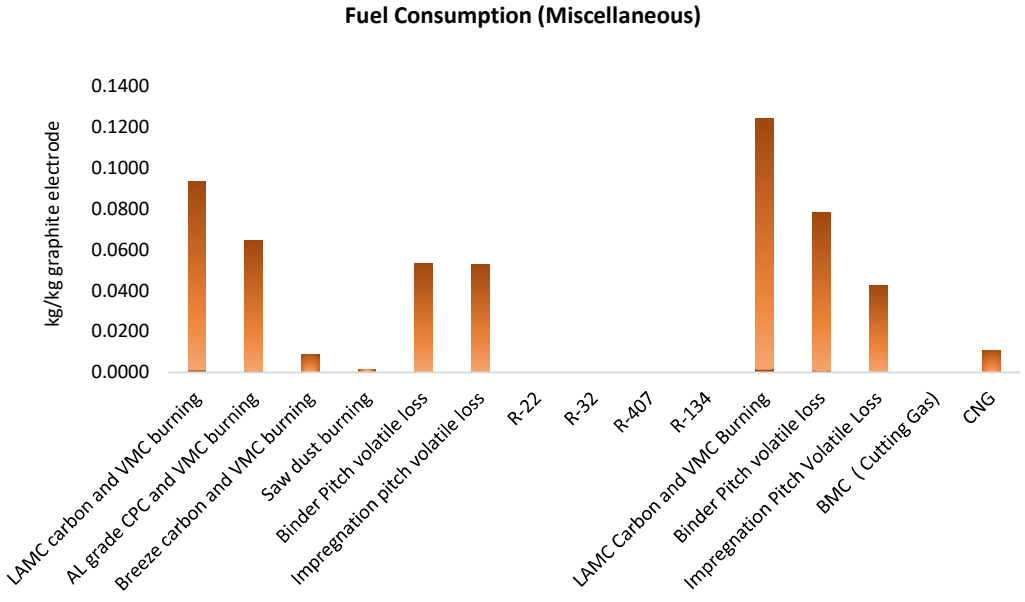


Fuel Consumption

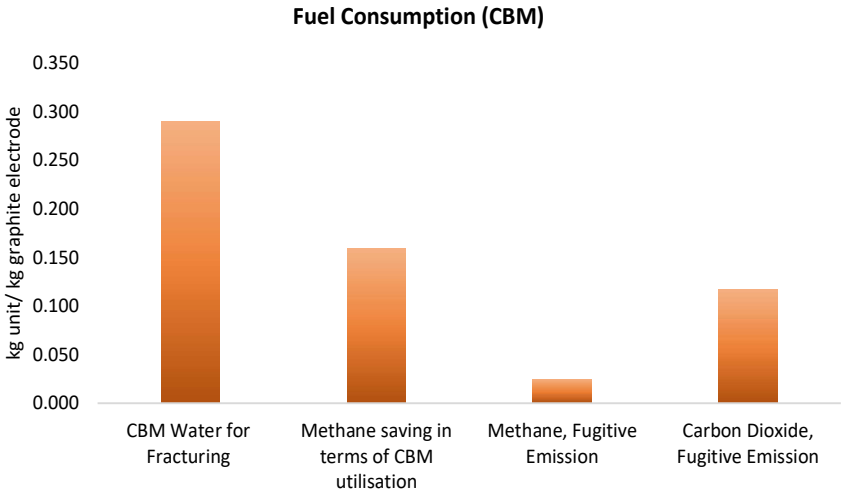
For LCI dataset, fuel consumption has been segregated for ease of inflow generation categorically into Diesel, CBM and Miscellaneous for Durgapur and Satpur inclusively. 36% of diesel is consumed by the machineries, 19% by both DG set and own/ controlled vehicles. 45% of diesel consumption is borne by utilized fuel in upstream supply chain. However, the consumption accounted with respect to the total production capacity for both GE division is minimal considering other inventory inflows. Density of diesel has been considered as 0.85 kg/l. Apart from that, in miscellaneous fuel category, 1.07% is extrusion oil and lubricants, 0.4% is light diesel oil, 1.054% is low sulfur heavy stock and 0.086% hydraulic oil. Unit wt. of hydraulic oil is considered as 0.869 kg/l, and 0.88 kg/l for LDO and extrusion oil. The liquid fuel consumption is comparatively much more lesser than the carbon based fuel consumption. 40% of LAMC carbon and VMC burning, 11.8% of Al grade CPC carbon and VMC burning, 24.2% of binder pitch volatile loss have been contemplated in the inflow boundaries in conjunction with Durgapur and Satpur division for 1 kg graphite electrode production. The significant divisions are depicted in the following figures.


Fuel Consumption (Diesel)
kg/kg graphite electrode
0.0025
0.002
0.0015
0.001
0.0005
0
Diesel in own/controlled vehicles
Diesel in own/controlled vechiles in project work
Diesel in own/controlled vehicles in ICD ware house
Diesel in DG Set
Diesel in Machinery
Diesel


Fuel Consumption (Miscellaneous)
kg/kg graphite electrode
0.0035
0.003
0.0025
0.002
0.0015
0.001
0.0005
0
Petrol in own vehicles
LPG in site
LPG in Ofice...
LPG at Guest House
LDO (Light Diesel Oil)
Furnace Oil
CO2 in process during...
Lubricants
Extrusion oil (lub oil)
Wood and paper in...
CO2 fire extingushers
Refrigerant (type R-22)
Refrigerant (type R-410)
Refrigerant (type R-407)
Refrigerant (type R-32)
Refrigerant (R-22) GH
Low Sulphur Heavy...
Hydraulic Oil
Low Sulphur Heavy...
LDO/ Indus.Diesel Oil
Extrusion Oil
Furnace Oil
LPG

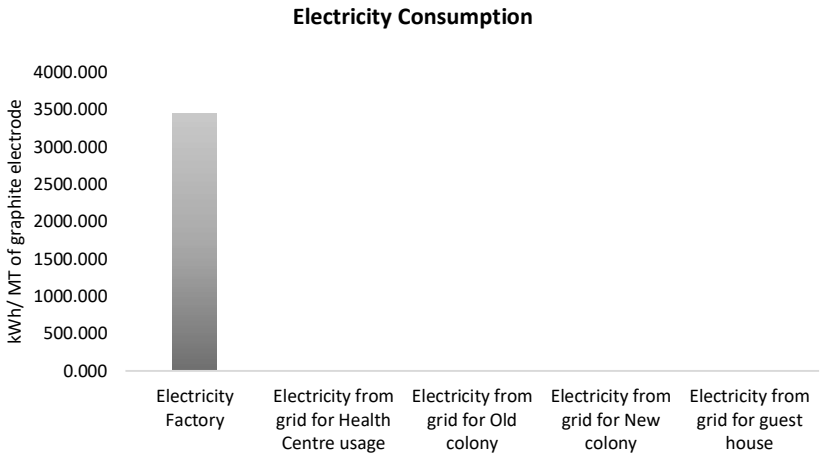

Fuel Consumption (Miscellaneous)
kg/kg graphite electrode
0.1400
0.1200
0.1000
0.0800
0.0600
0.0400
0.0200
0.0000
LAMC carbon and VMC burning
AL grade CPC and VMC burning
Breeze carbon and VMC burning
Saw dust burning
Binder Pitch volatile loss
Impregnation pitch volatile loss
R-22
R-32
R-407
R-134
LAMC Carbon and VMC Burning
Binder Pitch volatile loss
Impregnation Pitch Volatile Loss
BMC (Cutting Gas)
CNG

For coal bed methane, during its production phase, water is fractured and released into atmosphere. Water requirement for fracturing is considered to be 1.305 kg per cub.m of CBM produced. Along with that, fugitive emissions of fossil methane and carbon dioxide have been estimated during CBM capture, which is around 0.025 kg methane and 0.1170 kg carbon dioxide per kg of graphite electrode produced. Emission factor for CBM combustion is 2.75 kg CO2/kg of CBM and unit wt. is considered similar to natural gas i.e. 0.7168 kg/m^3.



Electricity Consumption

99% of electricity consumption is for factory units. The unit energy consumption is accounted as 3.445 kWh per kg of graphite electrode produced. Consideration of electricity consumption phase as well as the procurement and T&D loss phase in accordance with the Indian grid mix condition.



Water Consumption

The unit water consumption for 1 kg graphite electrode manufacturing is estimated as 1.071 litre.

Packaging & Finishing

For packaging and finishing stage, input flows like oil paint, wood and Metwrapp have been considered. The unit quantities are 0.00002 kg oil paint, 0.008 kg Metwrapp and 0.0000071 cub.m of wood for 1kg of graphite electrode production.

Impact Assessment Method

LCIA provides category indicators and selection of impact categories. There are many impact assessment methods like TRACI, commonly used in the United States; Ecoindicator, ReCiPe, IPCC and ILCD, employed in Europe; and CML method. CML is a procedure used to estimate the measure of environmental impact that is caused by the product. For this study, ecoinvent - CML v4.8 2016 have been used for assessing the environmental impact. The method is divided into baseline and non-baseline, the baseline being the most common impact categories used worldwide in LCA. The followings are the categories as depicted:

- Abiotic Depletion (ADP elements)
- Abiotic Depletion (ADP fossil)
- Acidification Potential (AP)
- Global Warming Potential (GWP 100 years)
- Eutrophication Potential (EP)
- Freshwater Aquatic Ecotoxicity Pot. (FAETP inf.)
- Marine water Aquatic Ecotoxicity Pot. (MAETP inf.)
- Terrestrial Ecotoxicity Pot. (TETP inf.)
- Human Toxicity Pot. (HTP inf.)
- Ozone Layer Depletion Potential (ODP, steady state)
- Photochemical Ozone Creation Potential (POCP)

Life Cycle Assessment Categories

Indicators	Indicators Abbreviation	Unit
Abiotic Depletion	ADP_{fossil}	[GJ]
Abiotic Depletion	$ADP_{elements}$	[kg Sb eq.]
Acidification Potential	AP	[kg SO2 eq.]
Eutrophication Potential	EP	[kg Phosphate eq.]
Freshwater Aquatic Ecotoxicity Potential	FAETP	[kg DCB eq.]
Global Warming Potential[1]	GWP	[tonnes CO2 eq.]
Global Warming Potential[1], excluding biogenic carbon	GWP_{excl}	[tonnes CO2 eq.]
Photochemical Ozone Creation Potential	POCP	[kg Ethene eq.]
Ozone Layer Depletion Potential	ODP[2]	[kg R11 eq.]
Primary energy demand from renewable and non-renewable resources	PED[3]	MJ

1. GWP 100 years 2. Steady state 3. Net calorific value

Impact Categories

Acidification Potential (AP)

Acid deposition, or "acid rain“, is the result of atmospheric water reacting with acidic gases like Sulphur dioxide (SO_2). When this rain occurs, it usually does so far from the gas's original source. Depending on the makeup of the landscape ecosystems, it impairs ecosystems to differing degrees. Nitrogen oxides (NO_x), Sulphur oxides (SO_x), and Ammonia (NH_3) are among the gases that contribute to acid deposition. The acidification potential is represented in kg SO_2 equivalent, which is the reference unit.

Climate Change & Global Warming Potential (GWP)

The term "climate change" refers to the variation in the Earth's surface temperature resulting from the greenhouse effect induced by the emission of greenhouse gases due to human activities. An increase in global temperature is predicted to bring about disease outbreaks, desertification, increasing sea levels, and climatic disruptions. One of the most significant environmental repercussions of economic activity is climate change, which is also one of the most challenging to manage due to its widespread nature. The most prevalent factor to express climate change is as Global Warming Potential over the time scale of different years, most commonly 100 years (GWP 100). The Global Warming Potential is represented with the reference unit of kg CO_2 equivalent

Abiotic Resource Depletion (ADP)

This impact category is referred to the unsustainable consumption of non-biological resources such as fossil fuels, minerals, metals, water, etc. The value of the abiotic resource consumption of a substance (e.g. lignite or coal) is a measure of the scarcity of a substance depending upon the amount of resource extracted. It is defined by the amount of resources that are depleted and characterized in antimony equivalents (kg Sb equivalent) for material resources and MJ of fossil fuels for energy use.

Ecotoxicity Potential

Environmental toxicity is measured as three separate impact sub-categories which inspect the freshwater, marine and land ecosystem. The emission of heavy metals, can have impacts on the ecosystem describing the fate, exposure and the environmental hazards. It is expressed using the reference unit, kg 1,4-dichlorobenzene equivalent (1,4-DB), and are measured separately for impacts of toxic substances on:
(i) Fresh-water aquatic ecosystems (FAETP)
(ii) Marine ecosystems (MAETP)
(iii) Terrestrial ecosystems (TETP)

Human Toxicity Potential (HTP)

The Human Toxicity Potential is an estimated index that reflects the potential hazard of a unit of chemical released into the environment, and it is based on both the inherent toxicity of a compound and its potential dose. Substances like arsenic, sodium dichromate, and hydrogen fluoride, are caused, for the most part, by electricity production from fossil sources. These are potentially dangerous chemicals to humans through inhalation, ingestion, and even contact. Cancer potency, for example, is an issue here. The reference unit to measure this impact category is in 1,4- dichlorobenzene equivalents (1,4-DB equivalent).

Impact Categories

Eutrophication Potential

Eutrophication is the accumulation of concentration of chemical nutrients in an ecosystem which leads to abnormal productivity. This causes excessive plant growth like algae in rivers which in turns causes severe reductions in water quality and animal populations affecting the ecosystem. Emissions of ammonia, nitrates, nitrogen oxides and phosphorous to air or water have an direct and indirect impact on eutrophication. It is expressed using the reference unit, kg PO_4^{3-} equivalents.

Ozone Layer Depletion Potential (ODP)

Ozone-depleting gases cause damage to stratospheric ozone or the "ozone layer". There is great uncertainty about the synergistic effects of different gases in the stratospheric zone, and all chlorinated and brominated compounds that are stable enough to reach the stratosphere can have an effect. CFCs, halons and HCFCs are the major causes of ozone depletion by increasing the amount of carcinogenic UVB light reaching the earth's surface. The ozone depletion potential is referred with the reference substance chlorofluorocarbon-11 (CFC-11), expressed in kg CFC-11 equivalent.

Photochemical Oxidant Formation Potential (POFP)

Ground-level ozone is enormously toxic to humans in high concentration in comparison to the stratospheric ozone. Photochemical ozone formation, also called "ground level ozone", is caused by the synergistic reaction of volatile organic compounds and nitrogen oxides in the presence of heat and sunlight. It depends largely on the amounts of Carbon monoxide (CO), Sulphur dioxide (SO_2), Nitrogen oxide (NO), ammonium and NMVOC (non-methane volatile organic compounds). expressed using the reference unit, kg ethylene (C_2H_4) equivalent.

Life Cycle Interpretation

The final stage of LCA, the interpretation of the impact assessment result is shown in the following Table.

Impact Categories	Method	Impact assessment result	Unit
acidification - acidification (incl. fate, average Europe total, A&B)	ecoinvent - CML v4.8 2016	**0.0260**	kg SO2-Eq
climate change - global warming potential (GWP100)	ecoinvent - CML v4.8 2016	**1.763**	kg CO2-Eq
ecotoxicity: freshwater - freshwater aquatic ecotoxicity (FAETP inf)	ecoinvent - CML v4.8 2016	**2.267**	kg 1,4-DCB-Eq
ecotoxicity: marine - marine aquatic ecotoxicity (MAETP inf)	ecoinvent - CML v4.8 2016	**10086.93**	kg 1,4-DCB-Eq
ecotoxicity: terrestrial - terrestrial ecotoxicity (TETP inf)	ecoinvent - CML v4.8 2016	**0.012**	kg 1,4-DCB-Eq
energy resources: non-renewable - abiotic depletion potential (ADP): fossil fuels	ecoinvent - CML v4.8 2016	**110.692**	MJ
eutrophication - eutrophication (fate not incl.)	ecoinvent - CML v4.8 2016	**0.015**	kg PO4-Eq
human toxicity - human toxicity (HTP inf)	ecoinvent - CML v4.8 2016	**4.424**	kg 1,4-DCB-Eq
material resources: metals/minerals - abiotic depletion potential (ADP): elements (ultimate reserves)	ecoinvent - CML v4.8 2016	**5.25E-07**	kg Sb-Eq
ozone depletion - ozone layer depletion (ODP steady state)	ecoinvent - CML v4.8 2016	**9.41E-08**	kg CFC-11-Eq
photochemical oxidant formation - photochemical oxidation (high NOx)	ecoinvent - CML v4.8 2016	**9.65E-04**	kg ethylene-Eq

Overview of Environmental Impacts

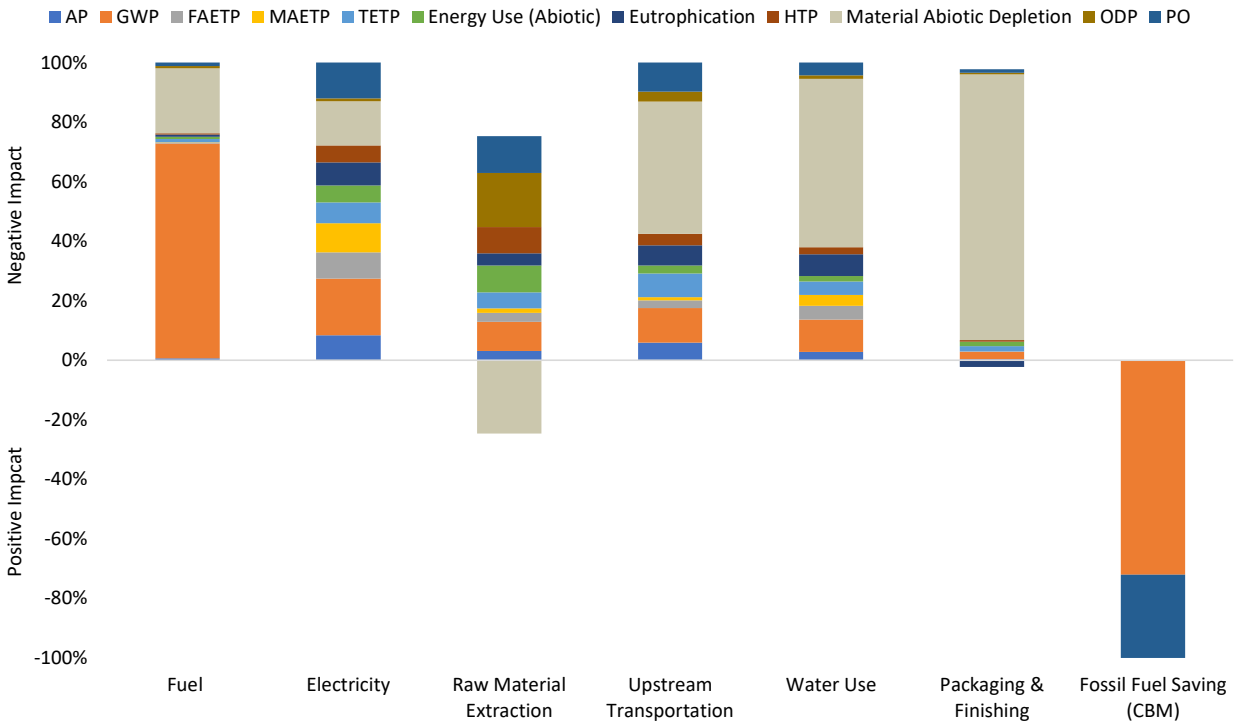


Fig. Contribution Analysis according to life cycle stages for cradle to gate approach

Life Cycle Stage wise Contribution Analysis

From the comprehensive insights of the impact assessment result, raw material extraction has major share of negative impact potential towards ozone depletion (51.8%), photochemical oxidant formation (48.5%) and global warming potential (46.96%). This is due to the life cycle impact of met coke production (13-52% of total raw material respectively) i.e. coking of coal life cycle. However, a net 118.78% of resource saving potential is the positive attributional impact element in regard to the sustainable consumption approach. However, the carbon footprint for total raw material extraction life cycle stage is found to be 0.828 kg CO_2 equivalent for 1 kg graphite electrode production. The usage of CPC, which is a petroleum byproduct tends to net savings in ADP for this stage which envisage sustainable consumption of waste product.

For upstream transportation stage, utilization of fuel and their life cycle impact contributes to a minimum share of net impact potential, quantitatively 0.58% - 26.9% of total impact for different impact indicators. However, ADP showed 26.9% share of impact due to utilization of fuel with the inclination of resource depletion, but the quantitative impact is very minimum consider to other impact categories.

Under the manufacturing stage, usage of electricity predominantly along with minimum fuel consumption, have been considered in the LCI due to which the impact potential of two prime energy inputs are analyzed separately. For fuel consumption impact, 0.28-23.77% share of impact is contributed to ODP, POFP, GWP, TETP, ADP: Energy Use. However, utilization of petroleum waste recovered during the fuel production life cycle saved 0.51% share impact of ADP material resource potential. But, the impact share of extrusion oil and lubricants showed a negative effect of 24.76% impact share in ADP.

The insight of impact potential for electricity consumption is important due to its maximum share (9.224% -179%) among others for certain indicators like AP, GWP, FAETP, MAETP, Eutrophication and POFP with the highest share for ADP material resource (179%) due to utilization of hard coal.

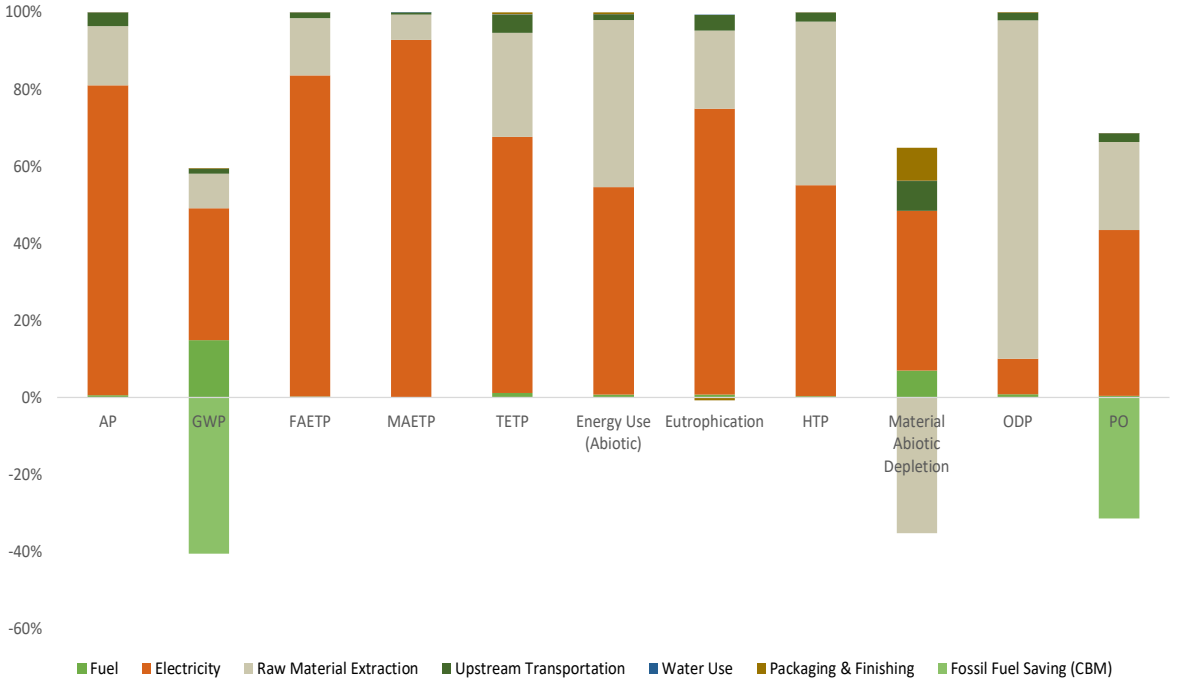


Fig. Indicator wise Contribution Analysis

For water usage, negligible impact potential has been observed with majority share 0.11% for ADP material resource for water consumption from water body.

For finishing and packaging stage, 28.82% impact share is observed for ADP material use. However, very minimum positive impact share of 0.704% is there for upcycling the package during their life cycle by contributing certain nitrogen compound in nature through eutrophication savings.

Throughout this life cycle assessment, sustainable consumption of CBM as a fuel not only saved natural resource like methane, but also saved a humongous amount of emission which could contribute towards GWP for its alarming impact potential than CO_2. Around 68% GHG emission reduction have been achieved for GWP potential and 45% for photochemical oxidant formation, This gives a insight on the product sustainability which confluence with the global climate risk mitigation strategy towards net zero goals.

Acidification Potential (AP)

Nitrogen oxides (NO_x), Sulphur oxides (SO_x), and Ammonia (NH_3) are among the gases that contribute to acid deposition. Estimated acidification potential for 1kg graphite electrode is 0.026 kg SO_2 equivalent. From the life cycle assessment insights, around 80.38% of acidification is causing from share of electricity production, 15.4% share is for raw material extraction and 0.67% impact share for fuel consumption that is negligible. LAMC has major share (5.146% out of 15.4%) causing acidification due to coking of coal. However, upstream transportation has 3.46% of impact potential for road transport out of which 99.7% allocation is for raw material transport. 2.38% share of acidification is estimated for raw material transport through sea water.

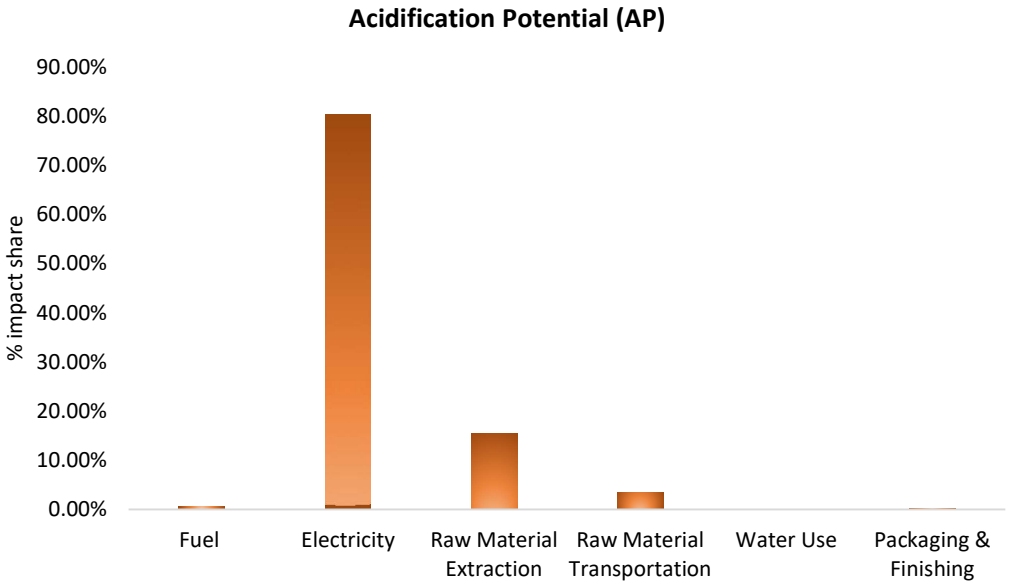


Fig. Contribution of different stages in Acidification Potential

The detail life cycle impact per kg of graphite electrode production for different stages are attributed in the following Table.

Life Cycle Inventory	Result	Unit
Fuel	1.75E-04	kg SO_2-Eq
Electricity	2.09E-02	kg SO2-Eq
Raw Material Extraction	4.01E-03	kg SO2-Eq
Upstream Transportation	9.00E-04	kg SO2-Eq
Water Use	1.37E-06	kg SO2-Eq
Packaging & Finishing	2.86E-05	kg SO2-Eq

Climate Change & Global Warming Potential (GWP)

Carbon dioxides (CO_2), Methane (CH_4) are major contributing gases to global temperature rise. Estimated global warming potential for 1kg graphite electrode manufacturing is 1.763 kg CO_2 equivalent. From the cradle to gate life cycle insights, around 3.163 kg CO_2 equivalent of GWP is causing from electricity production i.e. 179.44% share, 78.6% impact share for fuel consumption and 46.96% share is for raw material extraction. GWP impact of electricity shares into two phases, i.e., 78% of electricity for consumption in factory and 22% for procurement and transmission & distribution (T&D) loss stages. 75% of fuel impacts are aligned with utilization impact of CNG, LAMC carbon and VMC burning, binder pitch volatile loss etc. LAMC has major GWP share i.e. 13.73% followed by imported grade CPC i.e. 11.67% due to its production and logistic phase. However, upstream transportation has 5.25% of impact potential for road transport and 1.66% share of GWP is estimated for raw material transport through sea water. However, a substantial reduction of GWP by 68% is accounted for fossil methane savings through CBM utilization. This took forward to a substantial reduction of GWP by 3.75 kg CO_2 equivalent. The CBM utilization accounts for positive impact towards GWP reduction for the product life cycle.

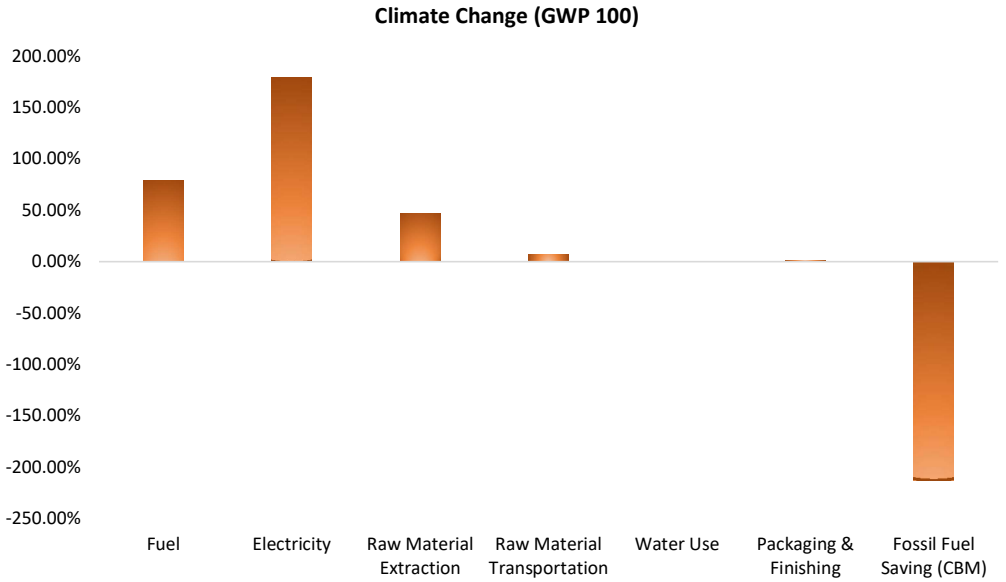


Fig. Contribution of different stages in Climate Change & Global Warming Potential

The stagewise life cycle impact on climate change per kg of graphite electrode production are represented in the following Table.

Life Cycle Inventory	Result	Unit
Fuel	1.39	kg CO2-Eq
Electricity	3.163	kg CO2-Eq
Raw Material Extraction	0.828	kg CO2-Eq
Upstream Transportation	0.122	kg CO2-Eq
Water Use	3.70E-04	kg CO2-Eq
Packaging & Finishing	1.43E-02	kg CO2-Eq
Fossil Fuel Saving (CBM)	-3.75	kg CO2-Eq

Fresh Water Aquatic Ecotoxicity (FAETP)

The release of heavy metals, can have impacts on the ecosystem describing the fate, exposure and the environmental hazards. The heavy metals like Nickel, Vanadium, Beryllium, Cobalt, Copper, Zinc and Selenium are the root causes for fresh water ecotoxicity. Estimated fresh water ecotoxicity potential for 1kg graphite electrode production is 2.266 kg 1,4 DB equivalent. From the comprehensive result, around 83.38% of ecotoxicity impact on fresh water have been found from share of electricity production i.e. 1.889 kg 1,4 DB equivalent, 0.283% of minor impact share for fuel consumption and 14.79% share of impact is for raw material extraction. LAMC has major share (7.79% out of 14.79%) causing ecotoxicity. However, upstream transportation only accounts for 1.5% ecotoxicity impact potential. However, estimated impact for water use is negligible.

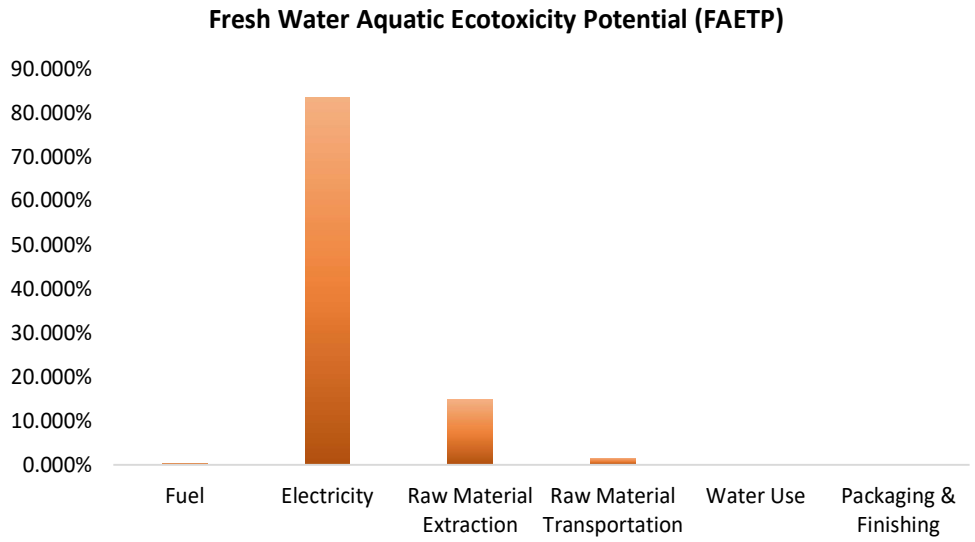


Fig. Contribution of different stages in Fresh Water Aquatic Ecotoxicity Potential

The stagewise life cycle impact on fresh water aquatic ecotoxicity per kg of graphite electrode production are represented in the following Table.

Life Cycle Inventory	Result	Unit
Fuel	6.41E-03	kg 1,4-DCB-Eq
Electricity	1.88973	kg 1,4-DCB-Eq
Raw Material Extraction	0.33518	kg 1,4-DCB-Eq
Upstream Transportation	0.03403	kg 1,4-DCB-Eq
Water Use	0.0002	kg 1,4-DCB-Eq
Packaging & Finishing	0.0013569	kg 1,4-DCB-Eq

Marine Aquatic Ecotoxicity (MAETP)

The gaseous and liquid emission of heavy metals, can have impacts on the marine aquatic ecosystem describing the fate, exposure and the environmental hazards. The heavy metals like Hydrogen fluoride, Nickel, Vanadium, Beryllium, Cobalt, and Selenium are the prime cause of impact on marine water ecotoxicity. Estimated marine ecotoxicity potential for 1kg graphite electrode production is 10,080 kg 1,4 DB equivalent. From the insights of LCA result, around 92.77% of ecotoxicity impact on marine ecosystem have been contributed by share of electricity production i.e. 9351.4 kg 1,4 DB equivalent and only 6.588% share of impact is for raw material extraction. Due to minimal fuel consumption per kg of graphite electrode production, the impact is negligible.

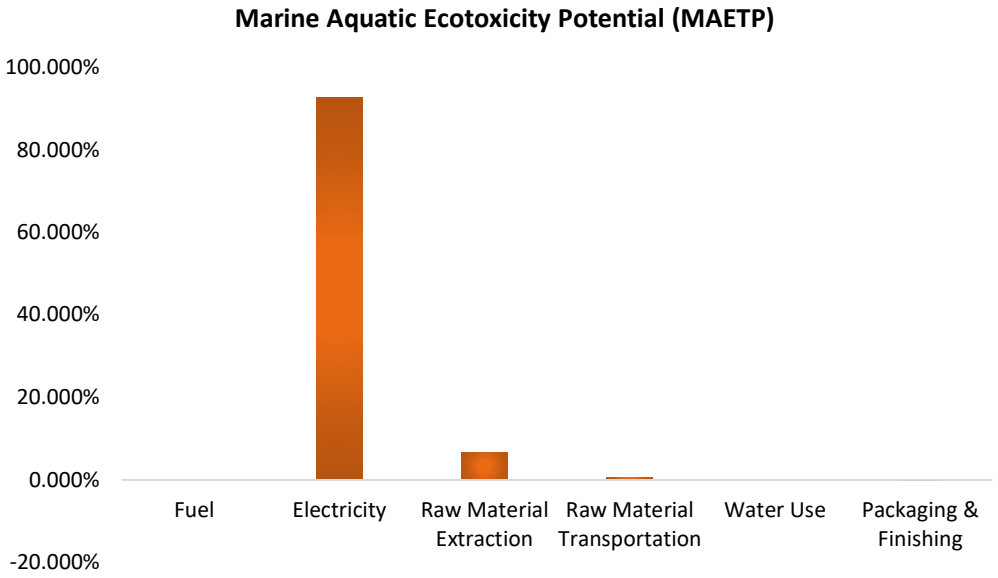


Fig. Contribution of different stages in Marine Aquatic Ecotoxicity Potential

The stagewise life cycle impact on marine aquatic ecotoxicity per kg of graphite electrode production are represented in the following Table.

Life Cycle Inventory	Result	Unit
Fuel	12.804	kg 1,4-DCB-Eq
Electricity	9351.4	kg 1,4-DCB-Eq
Raw Material Extraction	664.07	kg 1,4-DCB-Eq
Upstream Transportation	58.68	kg 1,4-DCB-Eq
Water Use	0.69	kg 1,4-DCB-Eq
Packaging & Finishing	-0.871	kg 1,4-DCB-Eq

Terrestrial Ecotoxicity (TETP)

The gaseous and liquid emission of heavy metals in air, water and land can have adverse impacts on the terrestrial ecosystem contemplating their hazard potential. The heavy metals like Mercury, Arsenic, Nickel, Vanadium, and Chromium are major contributor to terrestrial ecotoxicity impact. Estimated impact of terrestrial ecotoxicity potential for 1kg graphite electrode production is 0.012 kg 1,4 DB equivalent. Around 65.83% impact share for electricity consumption i.e. 0.079 kg 1,4 DB equivalent, 26.67% of ecotoxicity impact on terrestrial ecosystem have been contributed by share of raw material extraction. 4.75% impact share is responsible for upstream transportation, majorly for raw material transport through land and sea water.

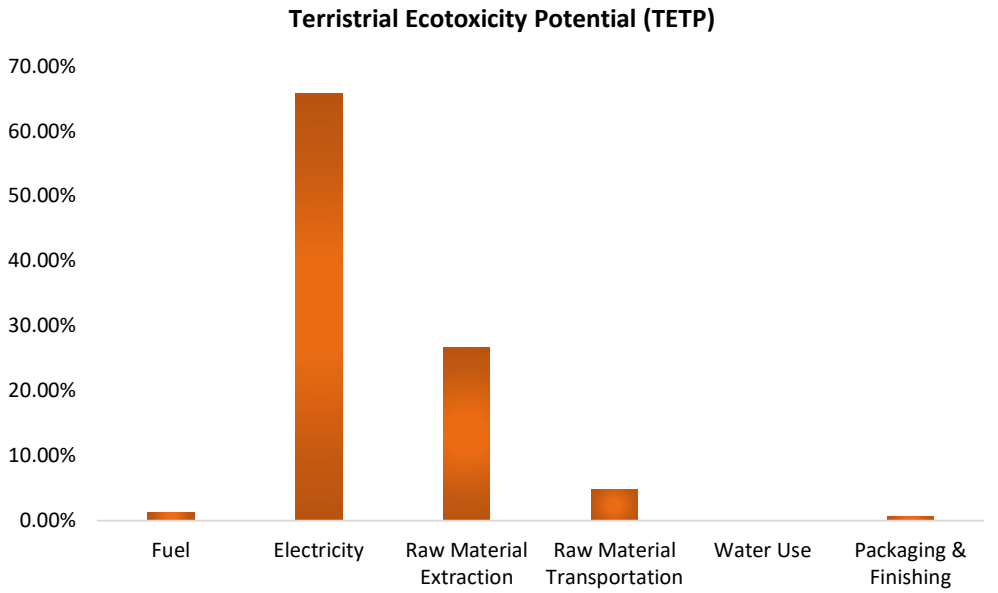


Fig. Contribution of different stages in Terrestrial Ecotoxicity Potential

The stagewise life cycle impact of terrestrial ecotoxicity per kg of graphite electrode production are represented in the following Table.

Inventory	Result	Unit
Fuel	1.44E-04	kg 1,4-DCB-Eq
Electricity	0.0079	kg 1,4-DCB-Eq
Raw Material Extraction	0.0032	kg 1,4-DCB-Eq
Upstream Transportation	0.00057	kg 1,4-DCB-Eq
Water Use	1.042E-06	kg 1,4-DCB-Eq
Packaging & Finishing	6.6231E-05	kg 1,4-DCB-Eq

Energy Resources: Abiotic Depletion Potential (ADP)

In context to the abiotic depletion with regard to the energy use, the depletion of fossil fuels and unsustainable consumption of abiotic resources are the root cause of this impact category. Estimated impact of abiotic depletion potential with regard to energy use for 1kg graphite electrode production is 110.69 MJ. Operational impact of hard coal, crude oil and natural gas in the product life cycle is associated with this impact potential. Around 53.79% impact share i.e. 59.54 MJ energy use potential is solely contributed by electricity consumption and 43.29% of impact i.e. 47.92 MJ is coupled with the raw material extraction in alliance with its life cycle production phase.

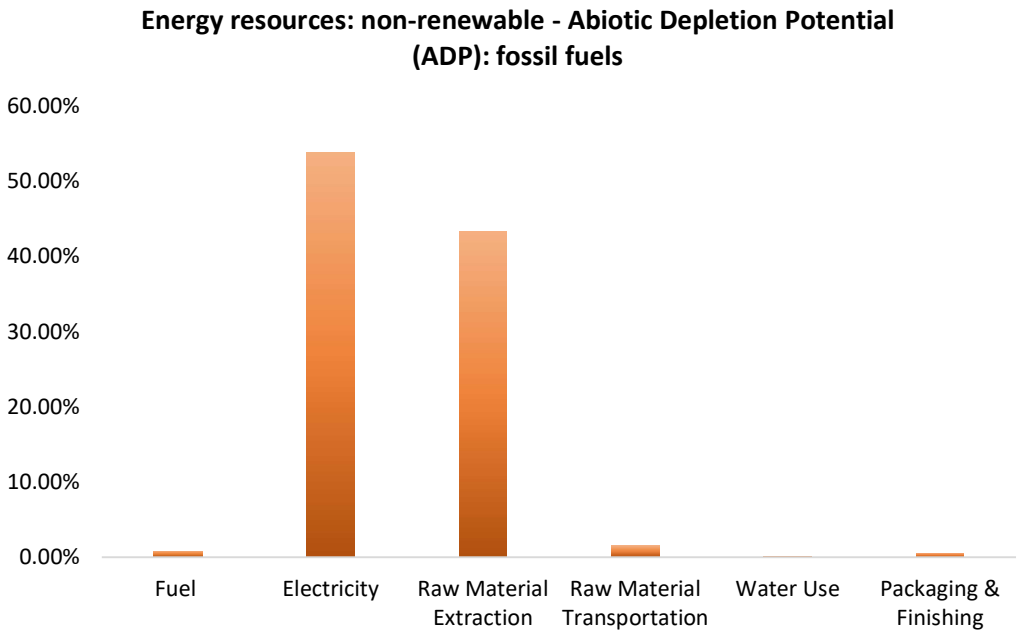


Fig. Contribution of different stages in Abiotic Depletion Potential: Energy Use

The stagewise life cycle impact of abiotic depletion energy use potential per kg of graphite electrode production are represented in the following Table.

Life Cycle Inventory	Result	Unit
Fuel	0.891	MJ
Electricity	59.54	MJ
Raw Material Extraction	47.92	MJ
Upstream Transportation	1.729	MJ
Water Use	0.004	MJ
Packaging & Finishing	0.557	MJ

Eutrophication

The accumulation of concentration of chemical nutrients like ammonia, nitrates, nitrogen oxides and phosphorous to air or water in an ecosystem causes Eutrophication. In context to this impact category, estimated impact of eutrophication for 1kg graphite electrode production is 0.015 kg PO_4^{3-} equivalent. From life cycle interpretation, operational impact of electricity production is associated by 73.33% i.e. 0.011 kg PO_4^{3-} equivalent and 20% of impact i.e. 0.003 kg PO_4^{3-} equivalent is associated with the raw material extraction phase. Emission of nitrogen and phosphate compounds in water occurs in accordance with hard coal and lignite mining activity as well as petroleum refining.

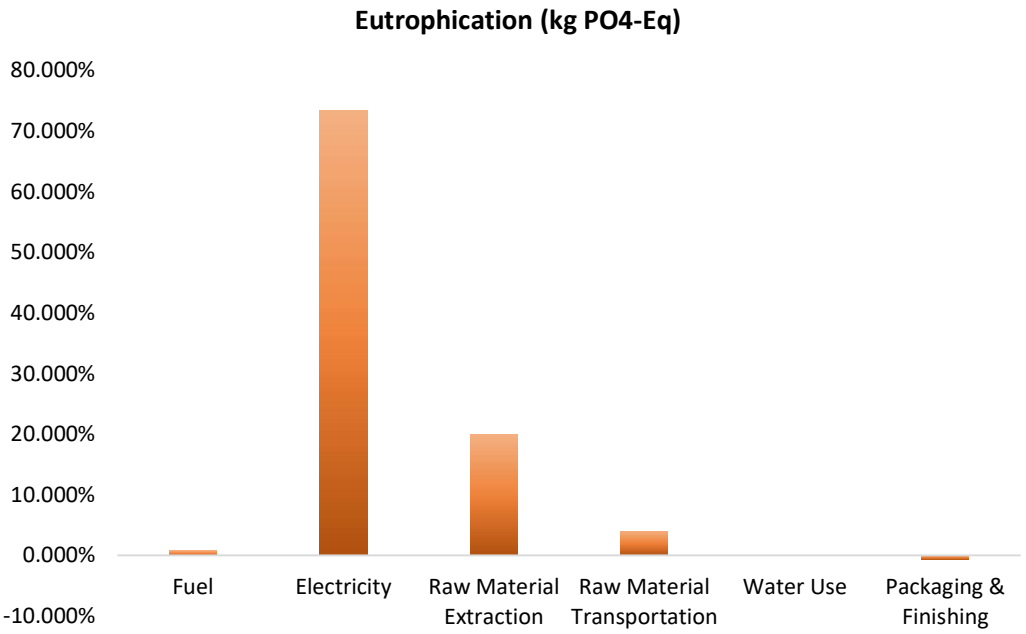


Fig. Contribution of different stages in Eutrophication Potential

The stagewise life cycle impact of eutrophication potential per kg of graphite electrode production are represented in the following Table.

Life Cycle Inventory	Result	Unit
Fuel	1.17E-04	kg PO4-Eq
Electricity	0.011	kg PO4-Eq
Raw Material Extraction	0.003	kg PO4-Eq
Raw Material Transportation	0.0006	kg PO4-Eq
Water Use	2.094E-06	kg PO4-Eq
Packaging & Finishing	-0.00010563	kg PO4-Eq

Human Toxicity Potential (HTP)

The gaseous and liquid emission of heavy metals in air, water and land can have adverse impacts on the human health considering their hazard potential. From life cycle analysis, Polycyclic aromatic hydrocarbons (PAH) and Benzene are the major contributor to human toxicity contributing more than 50% of impact. Estimated impact of human toxicity potential for 1kg graphite electrode production is 4.42 kg 1,4 DB equivalent. Around 42.2% of human toxicity impact by share was accounted for raw material extraction i.e. 1.866 kg 1,4 DB equivalent and 54.52% share of impact is for electricity production. 2.26% impact share is responsible for upstream transportation, majorly for raw material transport through land and sea water.

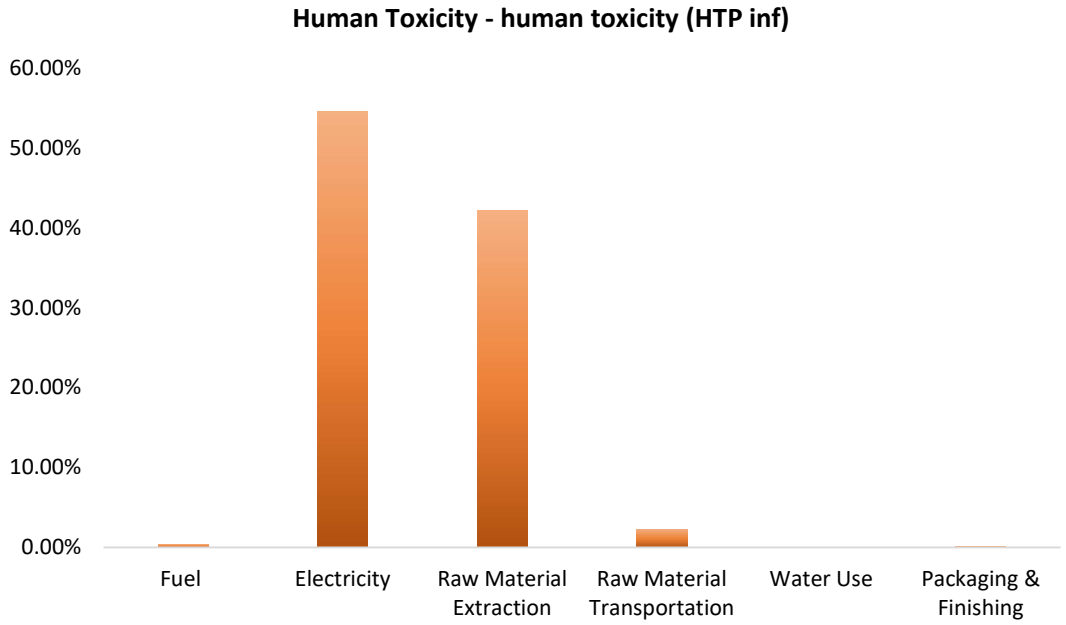


Fig. Contribution of different stages in Human Toxicity Potential

The stagewise life cycle impact of human toxicity per kg of graphite electrode production are represented in the following Table.

Life Cycle Inventory	Result	Unit
Fuel	1.87E-02	kg 1,4-DCB-Eq
Electricity	2.41	kg 1,4-DCB-Eq
Raw Material Extraction	1.866	kg 1,4-DCB-Eq
Raw Material Transportation	0.1	kg 1,4-DCB-Eq
Water Use	0.0002	kg 1,4-DCB-Eq
Packaging & Finishing	0.0075917	kg 1,4-DCB-Eq

Material Resources: Abiotic Depletion Potential (ADP)

In context to the abiotic depletion with regard to the material use, the depletion of fossil fuels and unsustainable consumption of abiotic resources are the root cause of this impact category. However, estimated impact of abiotic depletion potential with regard to material use for 1kg graphite electrode production is 5.25E-07 kg Sb-Eq which is net substantial. Operational impact of net resource savings during petroleum refinery waste recovery in the product life cycle is associated with this positive impact potential. Although, 140% impact share i.e. use is linked with the electricity production and 26.19% of impact i.e. 1.375E-07 kg Sb-Eq is coupled with the upstream transportation. But, due to petroleum waste recovery potential, net positive share of impacts are contributed by raw material extraction phase (118.78%) i.e. usage of CPC, a carbon based petroleum waste towards net resource savings.

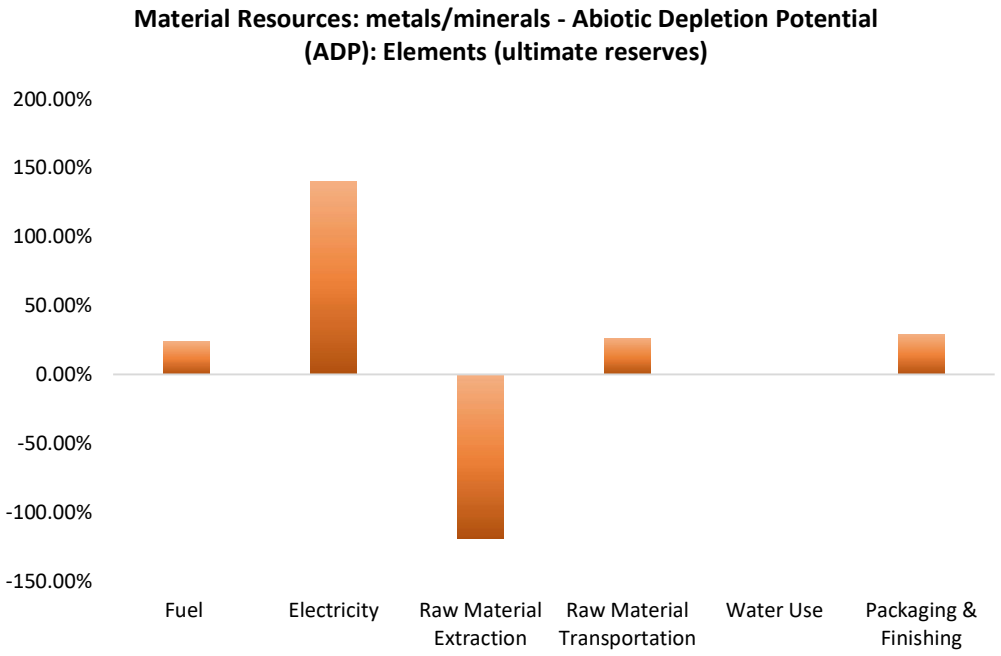


Fig. Contribution of different stages in Abiotic Depletion Potential, Resource Material

The stagewise life cycle impact of abiotic material resource depletion potential per kg of graphite electrode production are represented in the following Table.

Inventory	Result	Unit
Fuel	1.2477E-07	kg Sb-Eq
Electricity	7.35E-07	kg Sb-Eq
Raw Material Extraction	-6.236E-07	kg Sb-Eq
Raw Material Transportation	1.375E-07	kg Sb-Eq
Water Use	5.68E-10	kg Sb-Eq
Packaging & Finishing	1.5129E-07	kg Sb-Eq

Ozone Layer Depletion Potential (ODP)

The gaseous emission of CFCs, halons and HCFCs are the major causes of ozone depletion that have adverse impacts to the stratospheric ozone. From life cycle analysis, estimated impact of ozone depletion potential for 1kg graphite electrode production is 9.41E-08 kg CFC-11 equivalent. Around 87.57% of ODP impact by share was accounted of raw material extraction i.e. 8.241E-08 kg CFC-11 equivalent and 9.244% share of impact is for electricity life cycle. 1.94% impact share is responsible for upstream transportation, majorly for raw material transport through land and sea water.

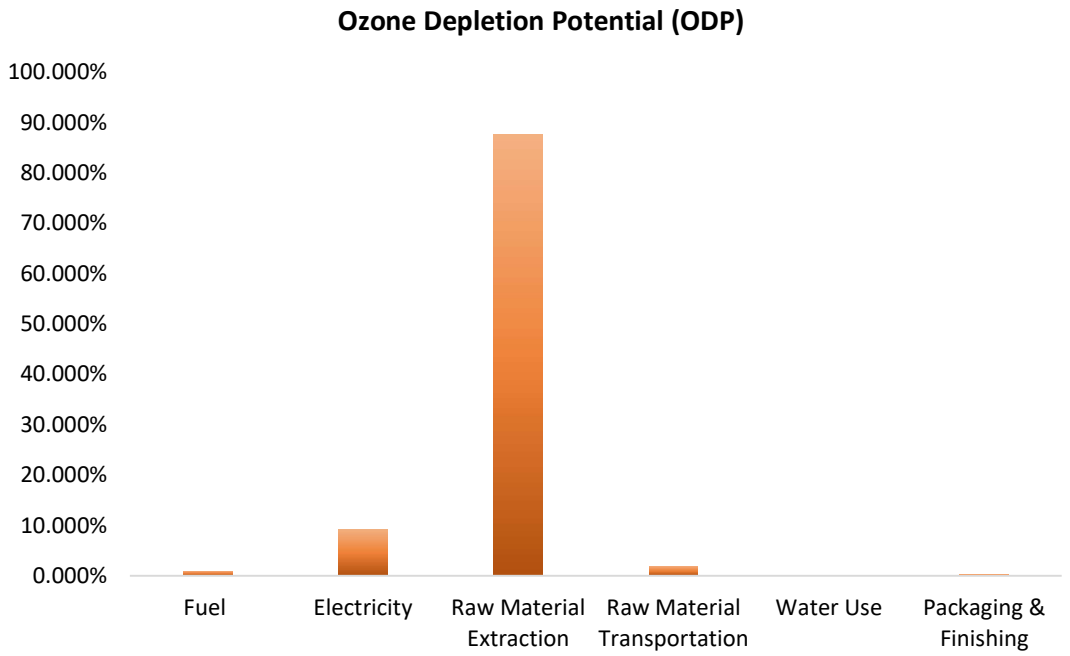


Fig. Contribution of different stages in Ozone Depletion Potential

The stagewise life cycle impact of ozone depletion potential per kg of graphite electrode production are represented in the following Table.

Inventory	Result	Unit
Fuel	8.3581E-10	kg CFC-11-Eq
Electricity	8.68E-09	kg CFC-11-Eq
Raw Material Extraction	8.241E-08	kg CFC-11-Eq
Raw Material Transportation	1.829E-09	kg CFC-11-Eq
Water Use	2.143E-12	kg CFC-11-Eq
Packaging & Finishing	1.6959E-10	kg CFC-11-Eq

Photochemical Oxidant Formation Potential (POFP)

The gaseous emission of Carbon monoxide (CO), Sulphur dioxide (SO_2), Nitrogen oxide (NO), and NMVOC (non-methane volatile organic compounds) have adverse impacts on photochemical ozone formation potential at ground level. From life cycle analysis, estimated impact of ozone depletion potential for 1kg graphite electrode production is 0.00097 kg ethylene-Eq. Around 113.4% of POFP impact by share was accounted for electricity consumption i.e. 1.10E-03 kg ethylene-Eq, 1.18% impact potential for fuel consumption and 60.15% share of impact is for raw material extraction life cycle. 5.76% impact share is responsible for upstream transportation, majorly for raw material transport through land and sea water. However, 82.47% fossil methane savings have been accounted for CBM utilization as fuel.

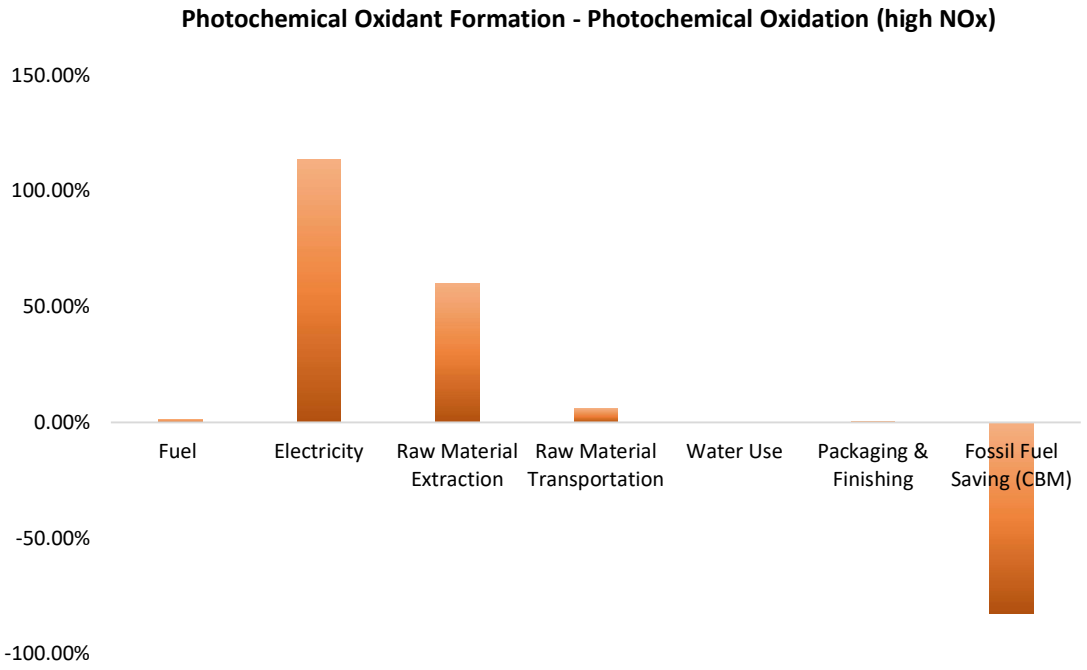


Fig. Contribution of different stages in Photochemical Oxidant Formation Potential

The stagewise life cycle impact of photochemical oxidant formation potential per kg of graphite electrode production are represented in the following Table.

Inventory	Result	Unit
Fuel	1.14E-05	kg ethylene-Eq
Electricity	1.10E-03	kg ethylene-Eq
Raw Material Extraction	5.84E-04	kg ethylene-Eq
Raw Material Transportation	5.59E-05	kg ethylene-Eq
Water Use	7.82E-08	kg ethylene-Eq
Packaging & Finishing	3.69E-06	kg ethylene-Eq
Fossil Fuel Saving (CBM)	-8.00E-04	kg ethylene-Eq

LCA Conclusions

The assessment of environmental performance of graphite electrode considering different operational stages including raw material, energy inputs, outputs, water usage, waste in their life cycle reflects on a first indication of environmental footprints as a starting reference point for further optimisation. It will help to further optimize the key hotspots of the operation phase to reduce the impact on the environment. The study focused on the following environmental indicators:

- Abiotic Depletion (ADP elements)
- Abiotic Depletion (ADP fossil)
- Acidification Potential (AP)
- Global Warming Potential (GWP 100 years)
- Eutrophication Potential (EP)
- Freshwater Aquatic Ecotoxicity Pot. (FAETP inf.)
- Marine water Aquatic Ecotoxicity Pot. (MAETP inf.)
- Terrestrial Ecotoxicity Pot. (TETP inf.)
- Human Toxicity Pot. (HTP inf.)
- Ozone Layer Depletion Potential (ODP, steady state)
- Photochemical Ozone Creation Potential (POCP)

As the results indicate, sustainable consumption of waste products from petroleum refinery as a raw material saves enormous footprint in terms of abiotic depletion of material elements by 118.78%. Besides, coal bed methane (CBM) utilization gives a major cognizance towards fossil methane savings and 68% GWP minimization potential for graphite electrode production. Grid electricity consumption gives an insight for major share of impact indicators like GWP, POFP, ODP, HTP, MAETP, FAETP etc. This gives an approach to strategies the consumption minimization of non-renewable energy sources. However, 80% of electricity consumption is confederated with the Durgapur GE division activity, only 20% electricity grid share is for Satpur division. Utilization CNG and CBM as sustainable fuels in the production chain ensembles an enormous footprint optimization and natural resource savings in terms of ADP impact share.

Appendices

In consideration with the production quantity of graphite electrodes, based on which this life cycle assessment was performed, is depicted as 85,254.078 MT for Durgapur GE division and 15,184 MT for Satpur, Nasik GE division for FY 2023-34. The calculations are based on the cumulative production quantity during the preparation of the inventories. These are represented in the following tables.

Computation of Raw Materials Inflow

Inputs	Quantity (FY 2023-24)	Unit	LCI Data	Inflow Unit	% Share
CPC, Regular Grade	29080.153	MT	0.290	kg/kg GE	24%
CPC, Imported Grade	36040	MT	0.359	kg/kg GE	30%
CPC, Al grade and LWG Pack Media	7850.057	MT	0.078	kg/kg GE	7%
Binder & PI Pitch	24662	MT	0.246	kg/kg GE	20%
LAMC	22926.353	MT	0.228	kg/kg GE	19%

Computation of Electricity Consumption Inflow

Inputs	Consumption (FY 2023-24)	Unit	LCI Data	Inflow Unit	Location
Electricity Factory	345862971	kWh	3443.544	kWh/MT	DGP + SP
Electricity from grid for Health Centre usage	3426.0	kWh	0.034	kWh/MT	DGP
Electricity from grid for Old colony	104641	kWh	1.042	kWh/MT	DGP
Electricity from grid for New colony	48984	kWh	0.488	kWh/MT	DGP
Electricity from grid for guest house	13423	kWh	0.134	kWh/MT	DGP
Total Electricity	346033445	kWh	3.445	kWh/kg	DGP + SP

Appendices

Computation of Fuel Inflow

Inputs	Consumption (FY 2023-24)	Unit	LCI Data	Inflow Unit
Petrol in own vehicles	7740	Litres	0.000061	kg/kg GE
LPG in site	456	Kg	0.0000045	kg/kg GE
LPG in Ofice (canteen,pantry,temple,lab)	551	Kg	0.0000055	kg/kg GE
LPG at Guest House	304	Kg	0.0000030	kg/kg GE
LDO (Light Diesel Oil)	0	Kg	0.0000000	kg/kg GE
Furnace Oil	0	Kg	0.0000000	kg/kg GE
CO2 in process during Project	0	kg	0.0000000	kg/kg GE
Lubricants	76080.6	Litres	0.0007	kg/kg GE
Extrusion oil (lub oil)	217.872	MT	0.0022	kg/kg GE
Wood and paper in ACHESON and LWG	174.166507	MT	0.0000025	m3/kg GE
CO2 fire extingushers	269.3	Kg	0.00000268	kg/kg GE
Refrigerant (type R-22)	26.4	Kg	0.00000026	kg/kg GE
Refrigerant (type R-410)	15.5	Kg	0.00000015	kg/kg GE
Refrigerant (type R-407)	34	Kg	0.00000034	kg/kg GE
Refrigerant (type R-32)	40.85	Kg	0.00000041	kg/kg GE
Refrigerant (R-22) GH	4.85	Kg	0.00000005	kg/kg GE
Low Sulphur Heavy Stock (LSHS)	245.49	MT	0.0024	kg/kg GE
CNG satpur	158855.902	MMBTU	0.01	MMBTu/kg GE
Hydraulic Oil	53970	L	0.0005	kg/kg GE
Oil Paint	1546	L	0.00002	kg/kg GE
Wood consumption	16426.31	CFT	0.0000046	m3/kg GE
Met wrap	160275	sqm.	0.008	kg/kg GE
Low Sulphur Heavy Stock (LSHS)	330	MT	0.0033	kg/kg GE
LDO/ Indus.Diesel Oil	250	KL	0.0022	kg/kg GE
Extrusion Oil	340	KL	0.0030	kg/kg GE
Furnace Oil	0	MT	0.0000	kg/kg GE
LPG	494	Kg	0.000005	kg/kg GE
Diesel in own/controlled vehicles	55888.02	L	0.00047	L/kg
Diesel in own/controlled vechiles in project work	3111.5	L	0.00003	L/kg
Diesel in own/controlled vehicles in ICD ware house	3384	L	0.00003	L/kg
Diesel in DG Set	52060.81176	L	0.00044	L/kg
Diesel in Machinery	219559	L	0.00186	L/kg
Diesel	271381	L	0.00230	L/kg

Appendices

Computation of Fuel Inflow (Satpur & Durgapur Additional)

Description	Consumption	Unit
LAMC carbon and VMC burning	0.0932	kg/kg GE
AL grade CPC and VMC burning	0.0643	kg/kg GE
Breeze carbon and VMC burning	0.0086	kg/kg GE
Saw dust burning	0.0014	kg/kg GE
Binder Pitch volatile loss	0.0535	kg/kg GE
Impregnation pitch volatile loss	0.0528	kg/kg GE
R-22	0.000000	kg/kg GE
R-32	0.000001	kg/kg GE
R-407	0.000002	kg/kg GE
R-134	0.000000	kg/kg GE
LAMC Carbon and VMC Burning	0.124355	kg/kg GE
Binder Pitch volatile loss	0.078010	kg/kg GE
Impregnation Pitch Volatile Loss	0.042691	kg/kg GE
BMC (Cutting Gas)	0.000084	kg/kg GE

Considerations:
Petrol unit wt. 0.79 kg/lt.
Lubricants unit wt. 0.88 kg/lt.
Wood, 1 ft3 = 0.0283168 m3, Unit wt. 700 kg/m^3,
https://matmatch.com/learn/property/density-of-wood.
Unit weight of Hydraulic oil is 0.869 kg/l,
https://www.crownoil.co.uk/products/lubricants/hydraulic-oil/hydraulic-.
Unit weight of Oil paint is 1.5kg/l,
https://justpaint.org/volume-weight-and-pigment-to-oil-ratios/
Polypropylene considered in ecoinvent for Metwrapp, unit wt. 946 kg/m^3
Extrusion Oil and LDO unit wt. 0.88 kg/lt.
Diesel Density considered 0.85 kg/L.

Appendices

Computation of CBM Inflow

Inputs	Consumption (FY 2023-24)	Unit	LCI Data	Inflow Unit
CBM_Fuel	8291988	kg	82.55821064	kg/MT
CBM-GAIL	436420	MMBtu	4.345	MMBtu/MT
Total CBM	860321.4944	MMBtu	8.566	MMBtu/MT
CBM Combustion Emission			0.438	kg CO2/kg GE
CBM Water for Fracturing	29139063.21	kg or lt	0.290	kg CBM water/kg GE
Methane saving in terms of CBM utilisation	860321.4944	MMBtu	0.1590	kg CBM/kg GE

Total CBM	860321.5	MMBtu	95% Methane consideration in CBM
Total CBM	22328784	m3	
Total CBM	222.3139	m3/MT	

1 US barrel = 119.24 litre Water		
Groundwater consumption	0.31	barrel/1000 ft3 CBM
	36.9644	lt/1000 ft3 CBM
	1.305	lt/m3 CBM
	1.305	kg/m3 CBM
	29139063.21	kg water
	0.290119682	kg water CBM/kg GE

Considerations:

1 MMBtu = 27.32 m^3

The water production as per Lawrence (1993) levelises the overall production rate to 0.31 barrels per 1,000 cubic-feet of coal bed methane.

Appendices

Computation of Upstream Transport Inflow (Durgapur)

Scope Emission	Inputs	LCI Data (FY 2023-24)	Unit
Scope 3 Upstream Transportation of Raw Material	Transport of CPC Regular Grade, Road	0.858	t.km/kg
	Transport of CPC Imported Grade, Road	0.183	t.km/kg
	Transport of CPC Imported Grade, Sea Freight	7.061	ship t.miles/kg
	Transport of CPC Al Grade & LWG Pack, Road	0.296	t.km/kg
	Transport of LAMC Met coke, Road	0.916	t.km/kg
	Transport of Binder Pitch, Road	0.158	t.km/kg
Scope 3 Upstream Transportation of Fuel	Transport of Low Sulfur Heavy Stock, Road	0.271	t.km/kg
	Transport of LDO, Road	0.271	t.km/kg
	Transport of Extrusion Oil, Road	0.271	t.km/kg
	Transport of Diesel, Road	0.003	t.km/kg
	Transport of LPG, Road	0.006	t.km/kg

Computation of Upstream Transport Inflow (Satpur)

Scope Emission	Inputs	LCI Data (FY 2023-24)	Unit
Scope 3 Upstream Transportation of Raw Material	Transport of CPC Regular Grade, Road	2.313	t.km/kg
	Transport of CPC Imported Grade, Road	0.228	t.km/kg
	Transport of CPC Imported Grade, Sea Freight	7.62	ship t.miles/kg
	Transport of CPC Al Grade, Road	1.847	t.km/kg
	Transport of Binder Pitch, Road	1.565	t.km/kg
Scope 3 Upstream Transportation of Fuel	Transport of Low Sulfur Heavy Stock, Road	0.271	t.km/kg
	Transport of Extrusion Oil, Road	0.271	t.km/kg
	Transport of Diesel, Road	0.271	t.km/kg

Appendices

Computation of Upstream Transport Inflow (Satpur & Durgapur)

Scope Emission	Inputs	LCI Data (FY 2023-24)	% Share	Inflow Unit
Scope 3 Upstream Transportation of Raw Material	Transport of CPC Regular Grade, Road	0.4631	52.1%	t.km/kg GE
	Transport of CPC Imported Grade, Road	0.0669	7.5%	t.km/kg GE
	Transport of CPC Imported Grade, Sea Freight	2.5488	100.0%	ship t.miles/kg GE
	Transport of CPC Al Grade & LWG Pack, Road	0.0362	4.1%	t.km/kg GE
	Transport of LAMC Met coke, Road	0.2091	23.5%	t.km/kg GE
	Transport of Binder Pitch, Road	0.1102	12.4%	t.km/kg GE
Scope 3 Upstream Transportation of Fuel	Transport of Low Sulfur Heavy Stock, Road	0.00089	0.1%	t.km/kg GE
	Transport of LDO, Road	0.0006	0.1%	t.km/kg GE
	Transport of Extrusion Oil, Road	0.0008	0.1%	t.km/kg GE
	Transport of Diesel, Road	0.000306496	0.0%	t.km/kg GE
	Transport of LPG, Road	2.95107E-08	0.0%	t.km/kg GE